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RESEARCH PAPER

Design of graded porous bone-like structures via a multi-material topology optimization approach

Zhi Zhao[1] · Xiaojia Shelly Zhang[1]

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Abstract

Graded porous structures combine robustness of porous structures and high stiffness of bulk designs. This study aims to design optimized graded porous bone-like structures through a novel multi-material topology optimization approach, which generalizes the concept of multiple materials. Namely, each material can have not only distinct material property but also a different level of local porosity, or a combination of both, thus allowing the realization of multiple levels of porosity. With separated density and material/porosity fields, we propose two types of multi-porosity local volume constraints to enable graded porosity considering linear and bi-linear material constitutive relations. Through the proposed framework, single- and multi-material structures can be obtained with a natural transition between the bulk and multiple levels of porous regions. We adopt the Bi-value Coding Parameterization (BCP) scheme combined with the Solid Isotropic Material with Penalization (SIMP) method to interpolate the stored energy functions. Through several examples with multiple porosity levels and various material properties, we demonstrate the effectiveness of the proposed framework with two novel constraints to generate optimized multi-material and multi-porosity structures. We further investigate the interactions among material properties, multiple porosity levels, structural stiffness, and robustness. Compared with conventional bulk designs, the optimized bone-like structures with multi-level graded porosity, although less stiff, are found to be more robust, i.e., their structural stiffness is less influenced by the load variations and material deficiency. The resulting graded porous composite designs showcase the capability of the proposed multi-material formulation to optimize the distributions of not only different types of materials but also multiple levels of porosity.

Keywords Graded porosity · Bone-like structure · Multi-material topology optimization · Bi-linear material · Local volume constraint

1 Introduction

Topology optimization is a powerful computational design tool used to find the optimal layouts of macro- and micro-structures. Given a design domain, it finds material distributions to minimize objective function values and fulfill a set of constraints. In the topology optimization field, the design of porous structures has gained interest over recent years. With the developments in additive manufacturing, fabricating the intricate geometry of the porous structures becomes possible (Wu et al. 2017; Liu et al. 2018a). Optimized porous structures are lightweight systems with robust performance and effective thermal/acoustic insulation functionality, making them promising candidates for applications including infill designs, buckling resilient reinforcements, and functional materials (see, e.g., Wu et al. 2017; Wang and Sigmund 2020; Vantyghem et al. 2019).

Porous structures are demonstrated to be robust with respect to load variations, material deficiency, and buckling compared to conventional bulk designs (Wu et al. 2017; Schmidt et al. 2019; Wang and Sigmund 2020). Yet, they typically have less stiffness than the bulk designs with the same amount of material used. Graded porous structures are promising because they combine the advantages of high robustness in porous structures and the high stiffness

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✉ Xiaojia Shelly Zhang
zhangxs@illinois.edu

[1] Civil and Environmental Engineering Department, University of Illinois at Urbana-Champaign, 205 North Mathews Ave, Urbana, IL, 61801, USA

in bulk designs. Many studies have proposed various strategies to design graded porous structures under either a single structural scale (Wu et al. 2017; Liu et al. 2018b; Schmidt et al. 2019; Groen and Sigmund 2018; Groen et al. 2019) or multiple scales (Zhang et al. 2019; Wu et al. 2019; Gao et al. 2019). For the studies on the single structural scale, two main streams are found in the literature, the homogenization-based approach and the density-based approach with optimization constraints. Compared with the homogenization-based approach, the latter has higher computational costs but circumvents the challenge of realizing smooth transitions among homogenized unit cells (Groen et al. 2019).

This study proposes a unique approach aiming at designing graded porous structures through a multi-material density-based topology optimization framework. Specifically, we generalize the concept of multiple materials, in which each material can have not only distinct material property (e.g., Young's modulus) but also a different level of local porosity, or a combination of both. To realize the design containing multiple levels of porosity, we propose two local constraints based on the pioneering work by Wu et al. (2017) and Schmidt et al. (2019). In addition, we separate the density field and the material/porosity field to facilitate the incorporation of proposed constraints. We interpolate the stored energy functions of multiple materials by employing the Bi-value Coding Parameterization (BCP) scheme (Bruyneel 2011; Gao et al. 2012) and the Solid Isotropic Material with Penalization (SIMP) method (Zhou and Rozvany 1991; Bendsøe and Sigmund 1999) with linear and bi-linear constitutive material models. Although the proposed framework and presented examples are demonstrated in two dimensions, we can readily extend the framework to three-dimensional (3D) cases, which have been successfully demonstrated in other studies (Schmidt et al. 2019; Wu et al. 2017; Xue et al. 2020a; Xue et al. 2020b; Geoffroy-Donders et al. 2020; Groen et al. 2020).

The remainder of the paper is organized as follows: Section 2 reviews relevant studies about porous structure design using topology optimization and related multi-material topology optimization frameworks. Section 3 presents the design parameterization, the interpolation scheme, and the proposed optimization formulation. Section 4 shows four examples to demonstrate the capabilities of the formulation and discuss the features of the obtained graded porous bone-like structures. Section 5 provides concluding remarks and future directions. The paper is complemented by three appendices. Appendix 1 presents the sensitivity analysis, Appendix 2 shows the algorithm and procedure of the proposed framework, and Appendix 3 outlines the nomenclature.

2 Related work

This section reviews prior studies that are relevant to this work, which include the design of porous structures and the multi-material topology optimization framework.

2.1 Topology optimization of porous structures

The porous structures can be designed on multiple scales or a single structural scale. The multi-scale optimization enlarges the design freedom by including the optimization of micro- and macro-structures, offering the flexibility to generate complex porous designs, and generally having higher computational cost compared to single-scale designs (Liu et al. 2018a). In the direction of multi-scale porous structure designs, Deng and Chen (2017) carried out a concurrent multi-scale topology optimization formulation to generate structures with multiple porous materials considering loading uncertainty. Similarly, Zhang et al. (2019) proposed a concurrent formulation to conduct multi-scale topology optimization to generate layer-wise graded lattice structures that outperform the uniform lattice structures in terms of stability. Gao et al. (2019) conducted a two-stage optimization (free material distribution and concurrent topology optimization) to design porous composites with multiple micro-structure units. Wu et al. (2019) developed an approximation of reduced substructure with penalization model to address the challenges regarding material connectivity and scale ratio when performing the homogenization-based multi-scale optimization to generate hierarchical lattice structures. Wadbro and Niu (2019) proposed a multi-scale topology optimization framework to generate coated structures with infill designs on the basis of Porous Anisotropic Material with Penalization linking up micro- and macro-structure.

The single structural scale topology optimization offers another way to generate porous structures. In general, this approach has lower computational costs and a smaller design space compared with the multi-scale optimization (Liu et al. 2018a). Several studies prescribe multiple unit cells in advance and project them into structures to enable a variety of porosities. For example, Liu et al. (2018b) adopted unit cells with multiple mechanical properties to optimize functionally graded materials and conducted compression experiments to test the behaviors of the optimized structures. Likewise, Xu et al. (2019) presented an isogeometric approach to create graded hierarchical lattice structures consisting of identical or varying unit cells based on multi-resolution topology optimization. Similarly, Zhao et al. (2019) integrated the optimization of relative unit density and the distribution of void regions

with two tailored filters to ensure minimum feature size and manufacturability when generating porous structures. Groen et al. (2019) introduced a homogenization-based optimization to construct coated structures with orthopedic infill. Wang and Sigmund (2020) conducted a series of buckling behavior simulations to compare stiffness and buckling response of structures with unit cells of a reference lattice and a topology optimized micro-structure. Additionally, graded porous structures can also be generated through parameter-based explicit geometrical methods on a single structural scale, for instance, Liu et al. (2020) developed a combined approach of the Moving Morphable Components and the Moving Morphable Voids to realize shell-graded-infill structures.

Imposing optimization constraints is an alternative approach to generate structural scale porous designs in the literature, which enables porous structures to be generated spontaneously without any prescribed unit cells. Wu et al. (2017) proposed a topology optimization formulation with localized material volume constraint to generate bone-like porous structures as lightweight infill. Following this initiative work, Schmidt et al. (2019) developed a new formulation with a modified local volume constraint capable of generating structures containing bulk and porous regions. Yi et al. (2019) incorporated the influence radius of the local volume constraint as an additional design variable to create functionally graded porous structures with a buckling constraint. In addition to the local volume constraint, Park et al. (2018) adopted a perimeter constraint to create complex bone internal structures, and Hu et al. (2019) developed a block-wise similarity constraint to generate structures guided by input exemplar images, which can be the shapes of porous structures.

This study aims to contribute to the field of designing graded porous structures with optimization constraints at the structural scale. We systematically design bone-like structures with multiple levels of porosity through a novel multi-material topology optimization approach, which generalizes each material to have not only distinct material property but also a different level of local porosity, or a combination of both. Based on the work by Wu et al. (2017) and Schmidt et al. (2019), we propose two local volume constraints to realize graded porous structure designs, which have a natural transition between the bulk design and multiple-level porous structures.

2.2 Multi-material topology optimization

Multi-material topology optimization enlarges the design space compared with single-material optimization. One key component of the multi-material topology optimization is to interpolate linear and nonlinear behaviors of various materials, which has been investigated widely. For example, Sigmund and Torquato (1997) proposed a three-phase multi-material interpolation scheme based on the SIMP method. Stegmann and Lund (2005) proposed the Discrete Material Optimization (DMO) method to effectively interpolate multiple material phases with the material mixing penalized. Zhou et al. (2018) introduced the Hypercube-to-Simplex Projection (HSP) method to interpolate materials fulfilling material unity constraint. In DMO and HSP methods, the number of design variables linearly depends on the number of materials; thus, the computational cost may be high when considerable materials are taken into account. To decrease the number of design variables, several studies propose new interpolation schemes. For instance, based on the quadrilateral shape function in the Finite Element Method (FEM), Bruyneel (2011) developed the Shape Function Parameterization (SFP) method to perform four-material interpolation with fewer design variables. Gao et al. (2012) proposed the Bi-value Coding Parameterization (BCP) method that generalizes the SFP method to handle an arbitrary number of materials. In addition, other studies developed interpolation schemes without introducing any new variables based on the peak function (Yin and Ananthasuresh 2001) or the ordered SIMP scheme (Zuo and Saitou 2017), both may require proper techniques for a stable transition between materials during optimization (Zuo and Saitou 2017). To consider multiple nonlinear materials, Zhang and Chi (2020) and Zhang et al. (2020) proposed a scheme that interpolates an arbitrary number of stored energy functions. In this study, we employ a BCP-type method to interpolate materials for its greatly reduced number of design variables (i.e., relatively low computational cost) without additional considerations on numerical stability. In addition, we make use of the scheme proposed by Zhang and Chi (2020) to interpolate multiple stored energy functions. We note that the proposed optimization formulation is general and can be directly applicable to other interpolation schemes.

3 Multi-material topology optimization for graded porous composite design

This section introduces the multi-material design parameterization and the proposed optimization formulation to design multi-porosity and multi-material structures.

3.1 Design parameterizations of multi-material graded porous structures

The overall design parameterization consists of representing multi-material graded porous structures with density and material/porosity variables as well as interpolating the stored energy functions of those composite structures with

multiple levels of porosity. This subsection describes these two components in detail.

3.1.1 Density and material/porosity parameterizations

In this work, we generalize the concept of material variables—each material variable can represent not only a different material constitutive relation but also a different level of local porosity (or both). We introduce both density and material/porosity variables to parametrize the designs containing several porosity levels. The density design variable, denoted by $\boldsymbol{\rho}$ (and ρ_e for element e), is filtered and projected to the physical density field, denoted by $\overline{\boldsymbol{\rho}}$ (and $\overline{\rho}_e$ for element e), which characterizes the spatial distribution of material. We propose $\xi_e^{(k)}$ to denote the kth ($k = 1, ..., N_\xi$) material/porosity design variable for element e. The material/porosity design variable $\xi_e^{(k)}$ is then filtered, projected, and mapped (using the BCP scheme) to the physical material/porosity field, $m_e^{(j)}$, which denotes the jth ($j = 1, ..., N_m$) physical material/porosity field for element e. The physical material/porosity fields can explicitly represent either different material types at each location over the design domain or various levels of porosity (or a combination of both). The different levels of porosity are controlled by the local volume constraints that are proposed in Section 3.2.2. We note that although the physical density field $\overline{\rho}_e$ and physical material/porosity fields $m_e^{(j)}$, $j = 1, ..., N_m$, are defined individually by two sets of design variables, they are coupled in the optimization formulation through multi-porosity local volume constraints (see Section 3.2.2) and multi-material stored energy interpolation scheme (see Section 3.1.2). As a result, their optimized distributions will collectively define the optimized design, as shown in Fig. 1, which presents an illustrative example to show the design parameterizations for generating multi-material graded porous structures.

Density parameterization includes filtering and projection from the density design variable to the physical density field. We first filter the density design variable to avoid mesh-dependent and checkerboard numerical issues (Sigmund 2007). The filtered density variable $\widetilde{\rho}_e$ is

$$\widetilde{\rho}_e = \frac{\sum_{i\in\mathscr{I}_e(r)} w_{i,e}\rho_e}{\sum_{i\in\mathscr{I}_e(r)} w_{i,e}}, \tag{1}$$

$$\mathscr{I}_e(r) = \{i : \|\boldsymbol{x}_i - \boldsymbol{x}_e\| \le r\}, \tag{2}$$

$$w_{i,e} = \max(1 - \frac{\|\boldsymbol{x}_i - \boldsymbol{x}_e\|}{r}, 0), \tag{3}$$

where $\mathscr{I}_e(r)$ is the elements in a circle with a radius of r, and $w_{i,e}$ is the weight that linearly depends on the distance of the element centroids $\boldsymbol{x}_i$ and $\boldsymbol{x}_e$. Then, we apply the Heaviside projection to the filtered density variable to enhance the discreteness of the optimized results (Wang et al. 2011). The projected physical density field $\overline{\rho}_e$ is

$$\overline{\rho}_e = \frac{\tanh(\beta_\rho\eta_\rho) + \tanh(\beta_\rho(\widetilde{\rho}_e - \eta_\rho))}{\tanh(\beta_\rho\eta_\rho) + \tanh(\beta_\rho(1 - \eta_\rho))}, \tag{4}$$

where β_ρ is the density projection parameter characterizing the discreteness of the projection, and η_ρ is the projection threshold.

Material/porosity parameterization includes filtering, projection, and mapping from the material/porosity design variable to the physical material/porosity fields. We first apply the distance-based filtering to obtain the kth filtered material/porosity variable $\widetilde{\xi}_e^{(k)}$, which is given as

$$\widetilde{\xi}_e^{(k)} = \frac{\sum_{i\in\mathscr{I}_e(r)} w_{i,e}\xi_e^{(k)}}{\sum_{i\in\mathscr{I}_e(r)} w_{i,e}}, \tag{5}$$

where $\mathscr{I}_e(r)$ and $w_{i,e}$ are the same in (1). Then, we use the Heaviside function to obtain the kth projected material/porosity variable $\overline{\xi}_e^{(k)}$, which is given as

$$\overline{\xi}_e^{(k)} = \frac{\tanh(\beta_\xi\eta_\xi) + \tanh(\beta_\xi(\widetilde{\xi}_e^{(k)} - \eta_\xi))}{\tanh(\beta_\xi\eta_\xi) + \tanh(\beta_\xi(1 - \eta_\xi))}, \tag{6}$$

where β_ξ is the projection parameter, and η_ξ is the threshold. We note that different from the purpose of the density projection, the projection on $\widetilde{\xi}_e^{(k)}$ is to reduce material/porosity mixing.

By employing the BCP scheme (Gao et al. 2012), we can use a small number of design variables to represent a larger number of physical material/porosity fields. Specifically, we map the kth projected material/porosity variable to the jth physical material/porosity fields, $m_e^{(j)}$, as

$$m_e^{(j)} = \frac{1}{2^{N_\xi}}\prod_{k=1}^{N_\xi}\left(1 + s_{j,k}\overline{\xi}_e^{(k)}\right), \quad j = 1, ..., N_m, \tag{7}$$

where N_m is the number of physical material/porosity fields, and $s_{j,k}$ is the mapping matrix, which is given by

$$s_{j,k} = \begin{cases} 1 & j \in [1, 2^{k-1}], \\ -1 & j \in [2^{k-1}+1, 2^k], \\ s_{t,k} & j \in [2^k+1, 2^{N_\xi}], \end{cases} \quad \text{where } t = 2^{\lceil \log_2 j \rceil} + 1 - j. \tag{8}$$

We remark that by construction, the unity condition, $m_e^{(1)} + m_e^{(2)} + ... + m_e^{(N_m)} = 1$, is satisfied and the number of physical material/porosity fields is $N_m = 2^{N_\xi}$. The number of material/porosity design variables is always fewer than the number of physical material/porosity fields, and thus can reduce the computational cost. In the case of considering a number of materials/porosities that is not a power of 2, we can divide one of the given materials into $2^{N_\xi} - N_m + 1$ portions so as to have $N_m = 2^{N_\xi}$.

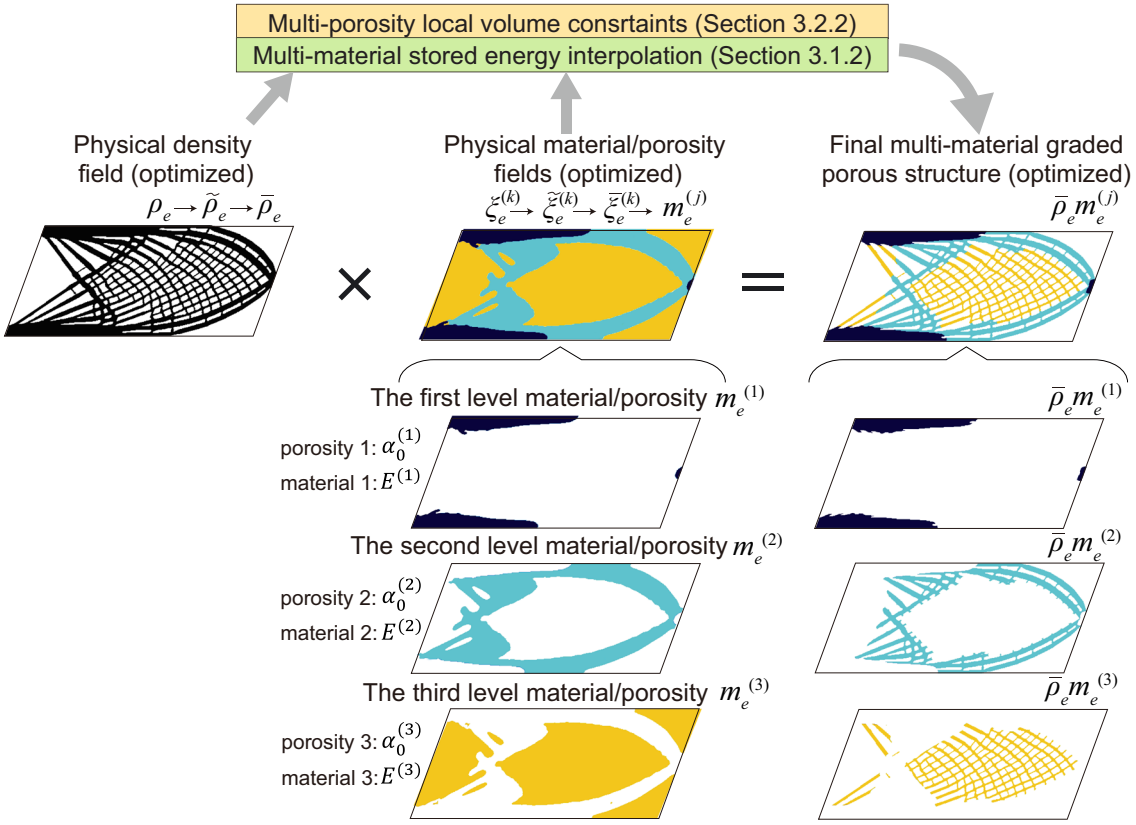


Fig. 1 Illustration of the design parameterizations generating a multi-material graded porous structure with density and material/porosity variables. Note that various colors represent different material properties and/or different porosity levels (or a combination of both). The optimized distributions of the physical density field $\overline{\rho}_e$ and physical material/porosity fields $m_e^{(j)}$, $j = 1, ..., N_m$, collectively define the optimized multi-material multi-level porosity design

As shown in Fig. 2, we use an example to further illustrate the mapping process from $\overline{\xi}_e^{(k)}$ to $m_e^{(j)}$ for element e. If two material/porosity design variables are applied, i.e., $N_\xi = 2$, we can represent four types of material/porosity, i.e., $N_m = 4$. The mapping functions from $\xi_e^{(k)}$ to $m_e^{(j)}$ are

$$
\begin{aligned}
m_e^{(1)} &= \left[\frac{1}{4}\left(1-\overline{\xi}_e^{(1)}\right)\left(1-\overline{\xi}_e^{(2)}\right)\right],\\
m_e^{(2)} &= \left[\frac{1}{4}\left(1+\overline{\xi}_e^{(1)}\right)\left(1-\overline{\xi}_e^{(2)}\right)\right],\\
m_e^{(3)} &= \left[\frac{1}{4}\left(1+\overline{\xi}_e^{(1)}\right)\left(1+\overline{\xi}_e^{(2)}\right)\right],\\
m_e^{(4)} &= \left[\frac{1}{4}\left(1-\overline{\xi}_e^{(1)}\right)\left(1+\overline{\xi}_e^{(2)}\right)\right].
\end{aligned}
\tag{9}
$$

Variable $m_e^{(j)}$ serves as a material/porosity indication variable for element e. The jth ($j = 1, ..., 4$) physical material/porosity field is selected for element e only when $m_e^{(j)} = 1$ and $m_e^{(\ell)} = 0, \ell \neq j$. Specifically, $\overline{\xi}_e^{(1)} = -1$ and $\overline{\xi}_e^{(2)} = -1$ (i.e., only $m_e^{(1)} = 1$ according to (9)) represent the 1st material/porosity is chosen for element e; $\overline{\xi}_e^{(1)} = 1$ and $\overline{\xi}_e^{(2)} = -1$ (i.e., only $m_e^{(2)} = 1$) represent the 2nd material/porosity is chosen for element e; $\overline{\xi}_e^{(1)} = 1$ and $\overline{\xi}_e^{(2)} = 1$ (i.e., only $m_e^{(3)} = 1$) represent the 3rd material/porosity is chosen for element e; and $\overline{\xi}_e^{(1)} = -1$ and $\overline{\xi}_e^{(2)} = 1$ (i.e., only $m_e^{(4)} = 1$) represent the 4th material/porosity is chosen for element e. In this example (with $N_\xi = 2$), the mapping functions happen to be the shape functions of the quadrilateral elements in FEM.

3.1.2 Multi-material interpolation of stored energy functions

We introduce the stored energy function of the employed material models and the corresponding multi-material interpolation scheme. This study applies isotropic linear and bi-linear constitutive material models (Ambartsumian 1966; Zhang and Chi 2020) under the assumptions of plane stress and small deformation in 2D. We present the stored energy function of the bi-linear material (which includes the linear material as a special case). We start from the kinematic relation, which is expressed by

$$
\boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2}\left((\nabla \boldsymbol{u})^T + (\nabla \boldsymbol{u})\right), \tag{10}
$$

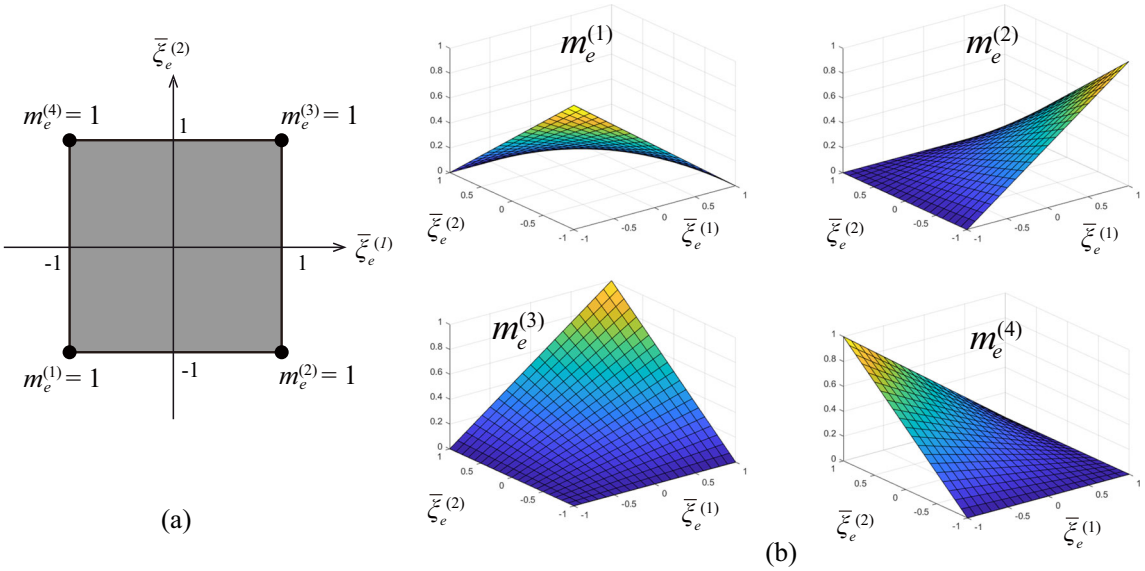


Fig. 2 Illustration of the mapping process from $\overline{\xi}_e^{(k)}$ $(k = 1, 2)$ to $m_e^{(j)}$ $(j = 1, ..., 4)$ for element e. (a) Feasible domain (i.e., gray region) of projected material/porosity design variables (after filtering and projection); (b) physical material/porosity field $m_e^{(j)}$ as a function of filtered and projected material/porosity design variable $\overline{\xi}_e^{(k)}$. Variable $m_e^{(j)}$ serves as a material/porosity indication variable for element e. The jth physical material/porosity field is selected only when $m_e^{(j)} = 1$ and $m_e^{(\ell)} = 0, \ell \neq j$

where $\boldsymbol{\varepsilon}$ is the strain, and $\boldsymbol{u}$ is the displacement. We define $\overline{\boldsymbol{\varepsilon}}$ (namely, $[\overline{\varepsilon}_1, \overline{\varepsilon}_2]^T$) as the principal strain, and $\overline{\boldsymbol{\sigma}}$ (namely, $[\overline{\sigma}_1, \overline{\sigma}_2]^T$) as the principal stress; therefore, the strain-stress relation can be expressed by

$$\begin{cases} \overline{\varepsilon}_1 = \frac{\overline{\sigma}_1}{E_t} - \frac{\nu_t}{E_t}\overline{\sigma}_2 & \text{if } \overline{\sigma}_1 \geq 0 \\ \overline{\varepsilon}_1 = \frac{\overline{\sigma}_1}{E_c} - \frac{\nu_c}{E_c}\overline{\sigma}_2 & \text{otherwise,} \end{cases}$$
$$\begin{cases} \overline{\varepsilon}_2 = \frac{\overline{\sigma}_2}{E_t} - \frac{\nu_t}{E_t}\overline{\sigma}_1 & \text{if } \overline{\sigma}_2 \geq 0 \\ \overline{\varepsilon}_2 = \frac{\overline{\sigma}_2}{E_c} - \frac{\nu_c}{E_c}\overline{\sigma}_1 & \text{otherwise,} \end{cases} \tag{11}$$

where E_t and E_c are Young's moduli for tension and compression, respectively; ν_t and ν_c are Poisson's ratios for tension and compression, respectively. As a result of thermodynamical consistency, the identity $\nu_t/E_t = \nu_t/E_t$ should be satisfied. Assuming $\overline{\varepsilon}_1 \geq \overline{\varepsilon}_2$ without the loss of generality, the stored energy function $W(\overline{\boldsymbol{\varepsilon}})$ is given as

$$W(\boldsymbol{u}) = \frac{1}{2}\overline{\boldsymbol{\varepsilon}}(\boldsymbol{u})^T \overline{\boldsymbol{D}}_\gamma(\overline{\boldsymbol{\varepsilon}}(\boldsymbol{u}))\overline{\boldsymbol{\varepsilon}}(\boldsymbol{u}),$$
$$\forall \overline{\boldsymbol{\varepsilon}}(\boldsymbol{u}) \in \mathcal{E}_\gamma \subset \mathbb{R}^2, \quad \gamma = 1, ..., 3, \tag{12}$$

where the strain condition $\mathcal{E}_\gamma$ and the tangent matrix $\overline{\boldsymbol{D}}_\gamma$ are given by

$$\mathcal{E}_1 = \begin{cases} \overline{\varepsilon}_1 + \nu_t\overline{\varepsilon}_2 \geq 0 \\ \nu_t\overline{\varepsilon}_1 + \overline{\varepsilon}_2 \geq 0 \end{cases}, \quad \mathcal{E}_2 = \begin{cases} \overline{\varepsilon}_1 + \nu_c\overline{\varepsilon}_2 \geq 0 \\ \nu_t\overline{\varepsilon}_1 + \overline{\varepsilon}_2 < 0 \end{cases},$$
$$\mathcal{E}_3 = \begin{cases} \overline{\varepsilon}_1 + \nu_c\overline{\varepsilon}_2 < 0 \\ \nu_c\overline{\varepsilon}_1 + \overline{\varepsilon}_2 < 0 \end{cases}, \tag{13}$$

and

$$\overline{\boldsymbol{D}}_1(\overline{\boldsymbol{\varepsilon}}) = \frac{E_t}{1-(\nu_t)^2}\begin{bmatrix} 1 & \nu_t \\ \nu_t & 1 \end{bmatrix},$$
$$\overline{\boldsymbol{D}}_2(\overline{\boldsymbol{\varepsilon}}) = \frac{1}{1-\nu_t\nu_c}\begin{bmatrix} E_t & E_t\nu_c \\ E_t\nu_c & E_c \end{bmatrix},$$
$$\overline{\boldsymbol{D}}_3(\overline{\boldsymbol{\varepsilon}}) = \frac{E_t}{1-(\nu_c)^2}\begin{bmatrix} 1 & \nu_c \\ \nu_c & 1 \end{bmatrix}, \tag{14}$$

respectively. In the case where the linear material is used, Young's moduli and Poisson's ratios for tension and compression are the same, i.e., $E_t = E_c = E$ and $\nu_t = \nu_c = \nu$. Under this condition, the tangent matrix reduces to a simplified expression, which is given as

$$W(\boldsymbol{u}) = \frac{1}{2}\overline{\boldsymbol{\varepsilon}}(\boldsymbol{u})^T\overline{\boldsymbol{D}}\overline{\boldsymbol{\varepsilon}}, \quad \forall \overline{\boldsymbol{\varepsilon}}(\boldsymbol{u}) \subset \mathbb{R}^2, \tag{15}$$

and $$\overline{\boldsymbol{D}} = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix}. \tag{16}$$

Having defined the material constitutive models, we interpolate their stored energy functions under the multi-material framework as follows (Zhang and Chi 2020):

$$\hat{W}(\overline{\rho}_e, m_e^{(1)}, ..., m_e^{(N_m)}, \boldsymbol{u}_h) = \epsilon_0 + (1-\epsilon_0)\overline{\rho}_e^{p_\rho}\sum_{j=1}^{N_m}(m_e^{(j)})^{p_m}W^{(j)}(\boldsymbol{u}_h), \tag{17}$$

where $\hat{W}(\cdot)$ is the interpolated stored energy function, ϵ_0 is a sufficiently small value to avoid numerical instability,

$W^{(j)}(\cdot)$ is the stored energy function for the jth material/porosity, p_ρ is the penalization parameter of the density variable, and p_m is the penalization parameter of the material/porosity variables. This material interpolation scheme is applicable for both linear and bi-linear materials.

3.2 Proposed topology optimization formulation

The general form of the proposed optimization formulation is given by

$$\begin{aligned}
&\min_{\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)}} J(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)},\boldsymbol{u}_h(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)}))\\
&\text{s.t. } 0<\rho_e\le 1,\quad e=1,...,N_e,\\
&-1\le \xi_e^{(k)}\le 1,\quad e=1,...,N_e,\quad k=1,...,N_\xi,\\
&G_G^{(\ell)}(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)})\le 0,\quad \ell=1,...,N_\ell,\\
&G_L(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)})\le 0,\\
&\text{with}\quad \boldsymbol{u}_h(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)})\\
&=\arg\min_{\boldsymbol{v}_h\in\mathcal{K}_h}\Pi_h(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)},\boldsymbol{v}_h).
\end{aligned}\tag{18}$$

where N_e is the number of elements, N_ξ is the number of the material/porosity design variables, and N_ℓ is the number of the global volume constraints. In the formulation, $J(\cdot)$ is the objective function, $G_G^{(\ell)}(\cdot)$ is the ℓth global volume constraint, $G_L(\cdot)$ is the multi-porosity local volume constraint, $\boldsymbol{v}_h$ is the displacement field within the admissible displacement space $\mathcal{K}_h$, and $\boldsymbol{u}_h$ is the equilibrating displacement field obtained from the minimization of the potential energy $\Pi_h(\cdot)$, which is given as

$$\begin{aligned}
&\Pi_h(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)},\boldsymbol{v}_h)\\
&=\sum_{e=1}^{N_e}\int_{\Omega_e}\hat{W}(\overline{\rho}_e,m_e^{(1)},...,m_e^{(N_m)},\boldsymbol{v}_h)d\Omega-\boldsymbol{F}^T\boldsymbol{v}_h,
\end{aligned}\tag{19}$$

where $\boldsymbol{F}$ is the external force, and Ω_e is the domain of element e. The mechanical analysis part follows the routine procedures in the density-based approach in topology optimization (Bendsoe and Sigmund 2013). The thin members constituting the porous structures are fully solved in the FE analysis (instead of being treated in a homogenized way). For the problems considering bi-linear materials, we employed Newton's method with inexact line search (Armijo 1966; Zhang et al. 2017). The tolerance used for Newton's method is 10^{-6}, and the maximum permissible Newton iteration is set to 20.

3.2.1 Global volume constraint

This study employs a flexible global volume constraint setting similar to the one proposed by Zhang et al. (2018). For materials with the same Young's modulus but distinct levels of porosity, we apply individual global volume constraints so as to drive the appearance of various levels of porosity in the design. Because with the same material behavior, the porosity tends to lower structural stiffness, without the individual global volume constraints, the compliance optimization framework tends to favor materials with fewer porosities (e.g., bulk design) to have higher structural stiffness. For cases with sufficiently distinguished material properties (e.g., bi-linear materials), a single global volume constraint is capable of generating graded porous structures. The global volume constraint $G_G^{(\ell)}$ is given as

$$\begin{aligned}
G_G^{(\ell)}(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},...,\boldsymbol{\xi}^{(N_\xi)}) &= \frac{\sum_{e=1}^{N_e}\sum_{j\in\mathcal{G}^\ell}\overline{\rho}_e m_e^{(j)}A_e}{\sum_{e=1}^{N_e}A_e}-V_0^{(\ell)}\\
&\le 0,\quad \ell=1,...,N_\ell,
\end{aligned}\tag{20}$$

where $V_0^{(\ell)}$ is the volume fraction of the ℓth constraint, $\mathcal{G}^\ell$ is the set of materials that belong to the ℓth constraint, A_e is the element area, and N_ℓ is the number of global constraints. When $N_\ell=1$, we assign one global volume constraint to all the materials/porosities. When $N_\ell=N_m$, we assign individual global volume constraint to each material/porosity type.

3.2.2 Proposed multi-porosity local volume constraints

The purpose of the proposed multi-porosity local volume constraints is to drive the forming of different levels of porosity. To devise the constraints, we first introduce a local volume fraction, $\hat{\rho}_e$, proposed in Wu et al. (2017), to represent the local material distribution within prescribed influence region $\mathscr{I}_e(R)$ with a radius of R for element e as follows:

$$\hat{\rho}_e(\boldsymbol{\rho})=\frac{\sum_{i\in\mathscr{I}_e(R)}\overline{\rho}_i A_i}{\sum_{i\in\mathscr{I}_e(R)}A_i},\tag{21}$$

where $\overline{\rho}_e$ is the physical density, and A_i is the element area. By constraining the upper bound of the local volume fraction (e.g., in the compliance minimization problems), porous structures have been realized with single materials (Wu et al. 2017; Schmidt et al. 2019). Here, a smaller radius R enables a stricter local constraint and results in a higher level of porosity (Wu et al. 2017). Note that the local volume fraction $\hat{\rho}_e$ takes into account all the materials/porosity levels within the influence region in the current study. Build upon this idea, we propose the following two new multi-porosity local volume constraints handling multiple materials assigned with different levels of porosity and/or distinct material properties.

The first multi-porosity local volume constraint (abbreviated as the L1 constraint hereafter) is inspired by the work of Schmidt et al. (2019). Similar to their study, we introduce a

non-uniform local volume fraction upper bound α_e to enable different levels of porous structures to be distributed in the design domain. However, we do not prescribe α_e in advance as in Schmidt et al. (2019); instead, we associate α_e with the physical material/porosity fields $m_e^{(j)}$ in order to correspond different levels of porosity to the physical fields $m_e^{(j)}$, $j = 1, ..., N_m$. We further introduce $\alpha_0^{(j)} \in (0, 1]$ as the prescribed local volume fraction upper bound that defines the jth porosity level. A lower magnitude of α_0 indicates a higher level of porosity, and $\alpha_0^{(j)} = 1$ indicates that jth level contains no porosity (bulk design). By assigning the various values of $\alpha_0^{(j)}$ in each material/porosity field, we can generate structures with different levels of porosity. Hence, we obtain the local volume fraction upper bound field $\alpha_e(\boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)})$, which is a function of physical material/porosity field and thus material/porosity variables, as follows:

$$\alpha_e(\boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)}) = \sum_{j=1}^{N_m} m_e^{(j)}(\boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)})\alpha_0^{(j)}. \tag{22}$$

Thus, the expression of the multi-porosity L1 constraint takes the following form:

$$G_{L1}\boldsymbol{\rho}, \boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)} = \left(\frac{1}{N_e}\sum_{e=1}^{N_e}\left(\frac{\hat{\rho}_e(\boldsymbol{\rho})}{\alpha_e(\boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)})}\right)^q\right)^{\frac{1}{q}} - 1 \le 0. \tag{23}$$

We use the p-norm function (Wu et al. 2017) with a parameter q to approximate the maximum ratio between local volume fraction and its upper bound field over the design domain.

The second multi-porosity local volume constraint (abbreviated as the L2 constraint hereafter) is based on a new concept of the local volume fraction defined for each individual material/porosity level, which is different from the first constraint (L1). For jth material/porosity, we define an individual influence region as $\mathscr{I}_e(R^{(j)})$ and introduce an individual local volume fraction for porosity level j, denoted by $\hat{V}_e^{(j)}$, to be associated with the corresponding physical material/porosity field $\boldsymbol{m}^{(j)}$ as follows:

$$\hat{V}_e^{(j)}(\boldsymbol{\rho}, \boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)}) = \frac{\sum_{i\in\mathscr{I}_e(R^{(j)})}\overline{\rho}_i m_i^{(j)}(\boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)})A_i}{\sum_{i\in\mathscr{I}_e(R^{(j)})}A_i}. \tag{24}$$

By associating influence region $\mathscr{I}_e(R^{(j)})$ with the material/porosity j, we can assign different influence regions (and influence radii) for various material/porosity fields.

Analogous to the ratio $\hat{\rho}_e/\alpha_e$ in (23), we introduce the ratio $\hat{V}_e^{(j)}/\alpha_0^{(j)}$ to enable the jth level of porosity. We sum over all the fractions in the constraint to restrict the material/porosity overlapping at each element. Similar to the L1 constraint, a p-norm approach is used. The multi-porosity L2 constraint, G_{L2}, is presented as

$$G_{L2}(\boldsymbol{\rho}, \boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)}) = \left(\frac{1}{N_e}\sum_{e=1}^{N_e}\left(\sum_{j=1}^{N_m}\left(\frac{\hat{V}_e^{(j)}(\boldsymbol{\rho}, \boldsymbol{\xi}^{(1)}, ..., \boldsymbol{\xi}^{(N_\xi)})}{\alpha_0^{(j)}}\right)\right)^q\right)^{\frac{1}{q}} - 1 \le 0. \tag{25}$$

The two multi-porosity local constraints, although with different forms, serve the same purpose of generating graded porous structures. However, note that the influence region $\mathscr{I}_e(R^{(j)})$ is associated with each material/porosity in the L2 constraint. Thus, the L2 constraint enables the assignment of various influence regions to control different types of materials and/or levels of porosity. Also, the influence radius should be reasonably larger than the element size in the finite element mesh. Therefore, lowering the influence radius requires a finer mesh resolution, which leads to a higher computational cost.

4 Numerical examples

This section presents four examples to demonstrate the effectiveness of the proposed formulation and the features of the obtained graded porous structures. Example 1 studies and compares the formulations with the L1 and L2 constraints to generate graded porous structures. Example 2 investigates the interactions among structural stiffness, multiple levels of porosity, and multiple material properties. Example 3 demonstrates the balance effect of the graded porous structures between porous and bulk designs. Example 4 presents a potential buckling resilient multi-material design enabled by the proposed formulation. Design domains, objectives, and material constitutive models of the four examples are presented in Table 1. In each example, the initial guesses for density and material/porosity design variables are $\rho_e = \sum_{\ell=1}^{N_\ell} V_0^{(\ell)}$ and $\xi_e^{(k)} = 0$, $k = 1, ..., N_\xi$, respectively, which represent uniform distributions for both density and material/porosity. Based on numerical experience, we apply two continuation strategies for parameters p_ρ, p_m, β_ρ, and β_ξ. In strategy 1, we sequentially increase the parameters as follows: $p_\rho = 1$, 1.5, 2, 2.5, 3 and then $p_m = 1$, 1.5, 2, 2.5, 3 at an interval of 100 optimization iterations. Then, the parameters β_ρ and β_ξ are doubled at an interval of 50 iterations until the results are sufficiently black-and-white. In strategy 2, $p_\rho = p_m = \beta_\rho = \beta_\xi = 1$ for the first 250 optimization iterations and increase simultaneously (with the same value ranges in strategy 1) at an interval of 100 iterations. From numerical investigations, strategy 1 is conservative and tends to reduce porosity mixing, which is used in Example 1, and strategy 2

Table 1 Design domains, objectives, and material constitutive models of the examples

Example	Design domain	Objective	Material constitutive model
1	Cantilever	$J_1(\cdot) = \boldsymbol{F}^T \boldsymbol{u}_h$	Linear
2	Disk	$J_1(\cdot) = \boldsymbol{F}^T \boldsymbol{u}_h$	Linear
3	Femur	$J_1(\cdot) = \boldsymbol{F}^T \boldsymbol{u}_h$ or $J_2(\cdot) = \sum_{i=1}^{N_c} \boldsymbol{F}_i^T \boldsymbol{u}_{h,i}$	Linear
4	Cantilever	$J_3(\cdot) = -\Pi_h$	Bi-linear

generates structures with less computational cost and is used for the rest of the examples. The optimization is terminated either when the change of the design variables is smaller than the tolerance value of 10^{-4} or reaching the maximum iteration number.

4.1 Example 1: Graded porous cantilever beam with two proposed local constraints

In example 1, we perform a total of 13 cases to compare structures with different levels of porosity generated by the two proposed multi-porosity local constraints. In particular, cases 1–7 compare the results obtained from the L1 and L2 constraints. Cases 8–12 compare the designs with various numbers of porosity levels and the corresponding objective function values. Case 13 showcases the capability of the L2 constraint to generate graded porous structures with different influence radii $R^{(j)}$. In this example, we use the same linear elastic material, i.e., $E = 1$. The design domain (discretized by 80,000 structured quadrilateral elements) and boundary conditions of the cantilever beam are shown in Fig. 3. Table 2 shows the parameters that are used in this example. For all cases, $r = 0.02$, $q = 10$, and each porosity level is assigned with an individual global volume constraint.

First, we demonstrate the two proposed local volume constraints by generating structures over a range of porosity levels (realized by assigning various values of $\alpha_0^{(j)}$ for $j = 1, ..., N_m$), and the corresponding optimized designs (cases 1–7) are shown in Fig. 4. Case 1 (Fig. 4a) is a bulk design ($\alpha_0^{(j)} = 1$, $j = N_m = 1$) without local volume constraints. Cases 2 and 3 generate identical one-level porous structures (Fig. 4b), because with one level of porosity assigned, the two proposed constraints (i.e., L1 and L2) become the same local volume constraint. In cases 4–7 (Fig. 4c–f), by assigning two and three different $\alpha_0^{(j)}$ values, we obtain two- and three-level porous structures, respectively. Based on the obtained designs, we observe that both proposed multi-porosity local volume constraints can successfully generate similar structures with various levels of porosity by controlling α_0. (The main difference of the L2 constraint with the L1 one is demonstrated in cases 9 and 13.) The individual global volume constraints remain inactive in the porosity cases; thus, the objective function values are not reported because the volume usage is different in each case. For the two three-level porous structures (in Fig. 4e, f) obtained from two different local volume constraints, Fig. 5 shows the distributions of $\overline{\boldsymbol{\rho}}\, \boldsymbol{m}^{(j)}$ associated with different $\alpha_0^{(j)}$ and the contours of the local volume fraction $\hat{\boldsymbol{\rho}}$ computed by (21). From the mechanics perspective, the outer boundary of a cantilever structure, particularly the regions of supports and the external load, undergo higher stress than the inner regions in general. Thus, this stress distribution guides the bulk design and different levels of porous structures to be formed hierarchically in the outer and inner regions, respectively.

Next, we aim to verify the effectiveness of the proposed optimization formulation to generate structures with various

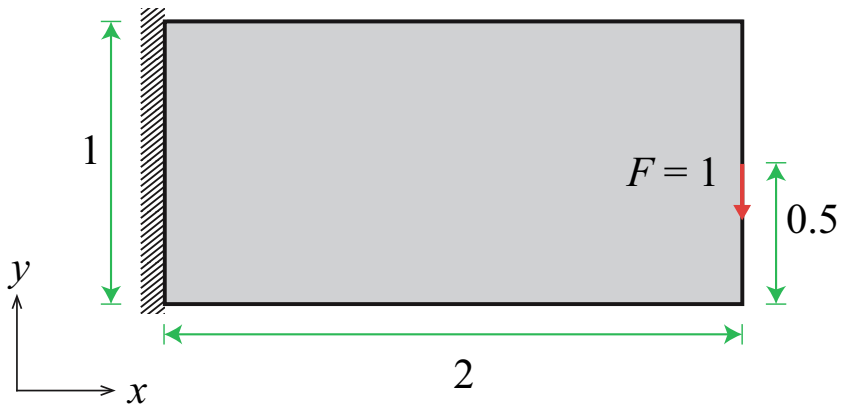


Fig. 3 Example 1: Design domain and boundary conditions

Table 2 Parameters of example 1, $\alpha_0^{(j)}$ and $R^{(j)}$, are the local volume fraction upper bound and the influence radius of the jth porosity, respectively, and $V_0^{(\ell)}$ is the volume fraction of ℓth global volume constraint

Case	# porosity levels	Local vol. constr.	$\alpha_0^{(j)}$	$R^{(j)}$	$V_0^{(\ell)}$	Final vol. fraction
1 (bulk)	–	–	[1]	[0.04]	[0.5]	0.5
2	1	L1	[0.4]	[0.04]	[0.5]	0.36
3	1	L2	[0.4]	[0.04]	[0.5]	0.36
4	2	L1	[1,0.4]	[0.04,0.04]	[0.25,0.25]	0.43
5	2	L2	[1,0.4]	[0.04,0.04]	[0.25,0.25]	0.48
6	3	L1	[1,0.7,0.4]	[0.04,0.04,0.04]	[0.1,0.2,0.2]	0.43
7	3	L2	[1,0.7,0.4]	[0.04,0.04,0.04]	[0.1,0.2,0.2]	0.48
8 (bulk)	–	–	[1]	[0.04]	[0.4]	0.4
9	1	L2	[0.5]	[0.04]	[0.4]	0.4
10	2	L2	[1,0.5]	[0.04,0.04]	[0.2,0.2]	0.4
11	3	L2	[1,0.75,0.5]	[0.04,0.04,0.04]	$[0.1\dot{3}, 0.1\dot{3}, 0.1\dot{3}]$	0.4
12	4	L2	$[1,0.8\dot{3},0.66\dot{6},0.5]$	[0.04,0.04,0.04,0.04]	[0.1,0.1,0.1,0.1]	0.4
13	2	L2	[0.5,0.5]	[0.04,0.12]	[0.2,0.2]	0.4

levels of porosity by assigning different values for $\alpha_0^{(j)}$ (local volume fraction upper bound) with respect to each material/porosity field j. To compare designs with different levels of porosity and their objective function values, we ensure global volume constraints are active in cases 8–12. Because the L1 and L2 constraints can produce similar results, we present results obtained from the L2 constraint. Figure 6 shows the objective function values and the corresponding topologies over a range of porosity levels. As expected, because the bulk design has higher stiffness compared with porous structures under the same material usage, the objective of the bulk design is the lower bound of all the obtained results. We observe that with more levels of porosity employed in the optimization, the objective function values are in a descending trend because the design space for porous structures is enlarged.

Finally, we demonstrate that the influence radius $R^{(j)}$ in the L2 constraint can be different for each porosity field, and thus, offers more design freedom to generate graded porous structures. This strategy cannot be achieved via the L1 constraint because R is associated with elements rather than individual porosity fields. We use one porosity field with $R^{(1)} = 0.04$ in case 9 and two porosity fields with $R^{(1)} = 0.12$ and $R^{(2)} = 0.04$ in case 13. We use $\alpha_0 = 0.5$ for both cases. As shown in Fig. 7, a one-level and a two-level porous structures are successfully obtained by employing different influence radius values. In particular, a higher radius value in case 13, $R^{(2)} = 0.12$, introduces a second (lower) level of porosity, as shown in Fig. 7b. Based on our numerical experience, with more than 2 levels of porosity incorporated, the results by controlling α_0 show smoother transition among different levels of porosity, therefore, we control α_0 to generate graded porous structures in the remaining examples.

4.2 Example 2: Multi-material graded porous disk design with various elastic properties

Example 2 investigates the interplay among the material properties, porosity levels, and the structural stiffness for the multi-material graded porous structures. We also aim to demonstrate that the proposed multi-material formulation can not only generate multiple levels of porous structures but also consider multiple material properties. Thus, for each material/porosity field, in addition to assigning various $\alpha_0^{(j)}$ values (i.e., $\alpha_0^{(j)} = 0.6, 0.7, 0.8$) to realize multiple porosity levels, we also use different linear materials with $E^{(j)} = 1, 0.75, 0.5$. The design domain, which is discretized by 80,000 polygonal elements (Talischi et al. 2012), and boundary conditions of the disk are shown in Fig. 8. Table 3 presents the parameters that are used in this example. For all cases, $r = 0.01$, $R = 0.03$, $q = 25$, and each material/porosity level is assigned with an individual global volume constraint. We present results obtained from the L1 constraint here.

Figure 9 shows the optimized results, all with three levels of porosity. Because the setup for the considered three cases is not for compliance comparison, the final objective function values are provided in Table 3 for reference purposes only. In case 1 (Fig. 9a), we assign uniform $\alpha_0^{(j)} = 0.6$ and use three materials with descending $E^{(j)}$ (j =1, 2, 3). While three materials are associated with the same

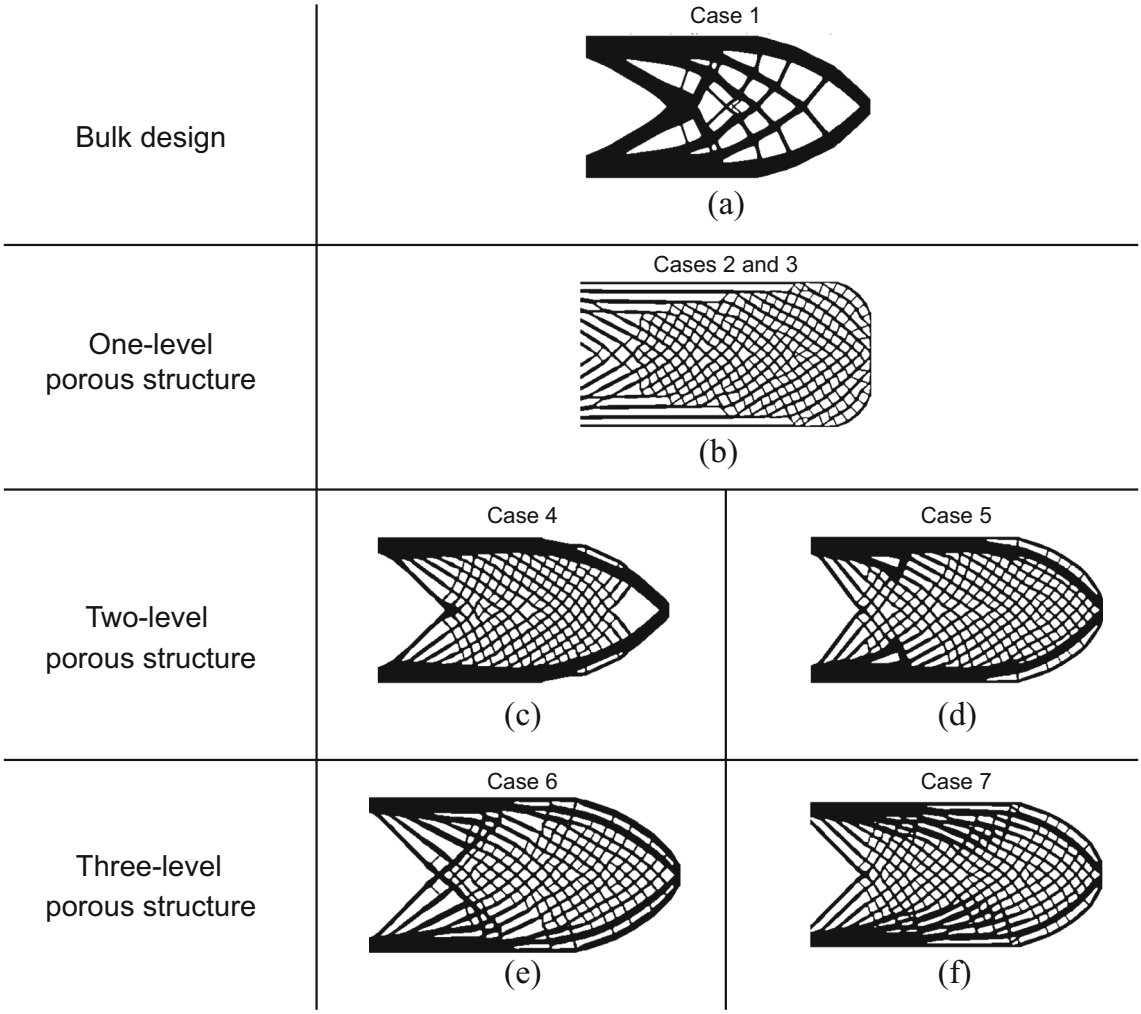


Fig. 4 Example 1: Final topologies of **a** bulk design; **b** one-level porous structure using the L1 and L2 constraints; **c** two-level porous structure using the L1 constraint; **d** two-level porous structure using the L2 constraint; **e** three-level porous structure using the L1 constraint; **f** three-level porous structure using the L2 constraint

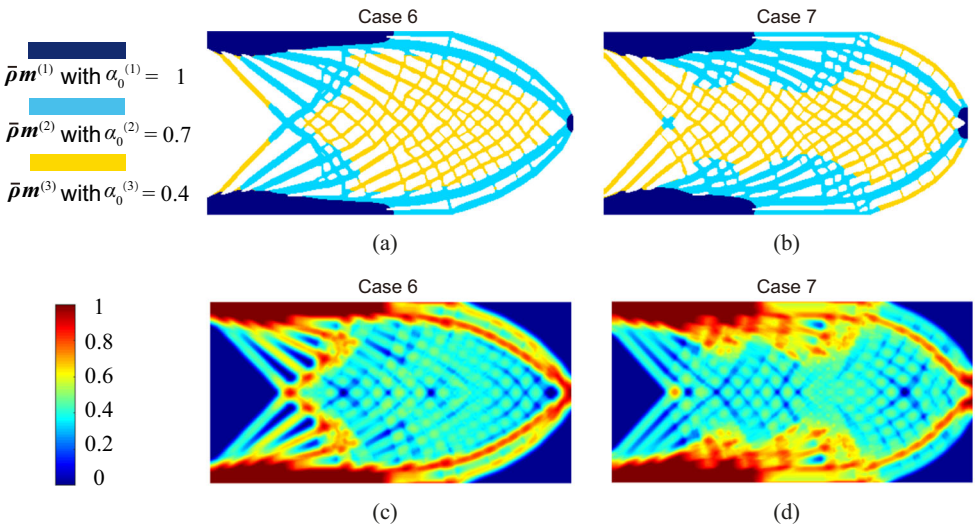


Fig. 5 Example 1: Three-level porous structures. Distributions of $\bar{\rho}\,\boldsymbol{m}^{(j)}$ using (a) the L1 constraint and (b) the L2 constraint. Contours of the local volume fraction $\hat{\rho}$ using (c) the L1 constraint and (d) the L2 constraint

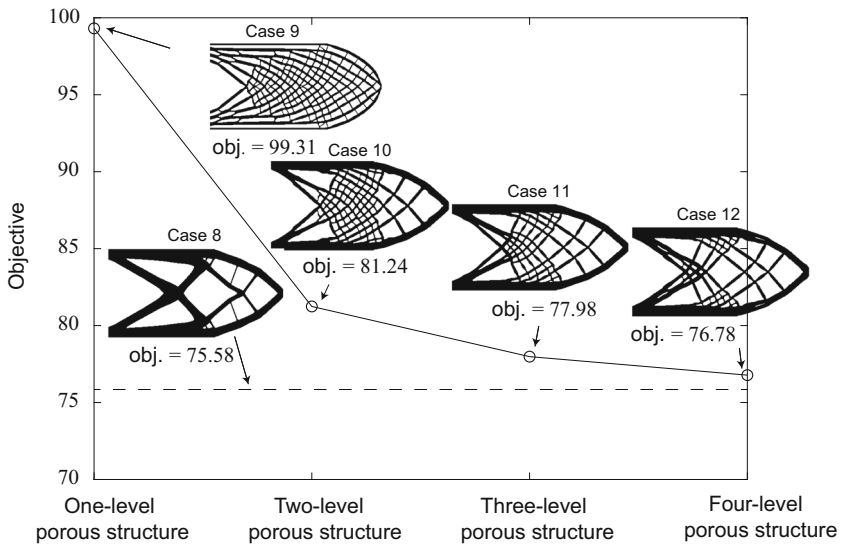


Fig. 6 Example 1: The objective function values versus the number of porous levels (all designs use the same volume fraction). By varying the number of porous levels, we can obtain a wide spectrum of designs ranging from bulk to porous structures

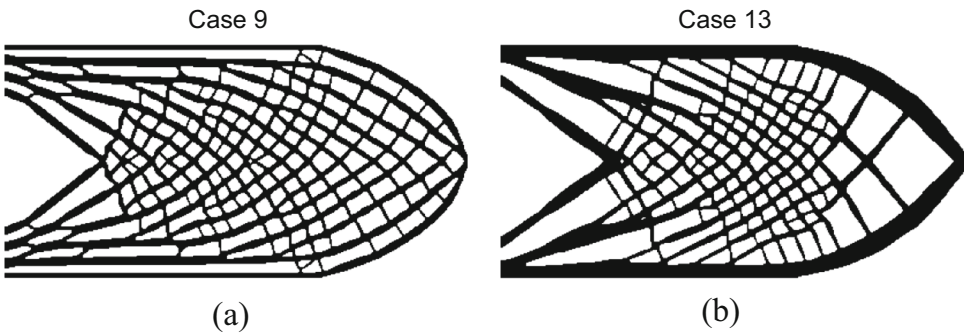


Fig. 7 Example 1: Final topologies of (a) one-level porous structure using the L2 constraint with $R^{(1)} = 0.04$ ($J^* = 99.31$); (b) two-level graded porous structure using the L2 constraint with $R^{(1)} = 0.04$ and $R^{(2)} = 0.12$ ($J^* = 89.72$)

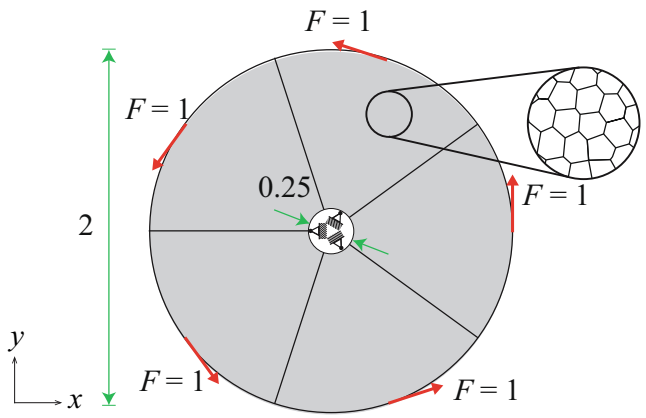


Fig. 8 Example 2: Design domain (discretized by polygonal elements) and boundary conditions

Table 3 Parameters of example 2, $\alpha_0^{(j)}$ is the local volume fraction upper bound of the jth porosity, $V_0^{(\ell)}$ is the volume fraction of ℓth global volume constraint, and $E^{(j)}$ is Young's Modulus of the jth material

Case	# material/porosity levels	Local vol. constr.	$\alpha_0^{(j)}$	$V_0^{(\ell)}$	$E^{(j)}$	Final vol. fraction	Final obj.	J^*
1	3	L1	[0.6,0.6,0.6]	[0.1,0.15,0.2]	[1,0.75,0.5]	0.45	795.32	
2	3	L1	[0.8,0.7,0.6]	[0.1,0.15,0.2]	[1,0.75,0.5]	0.45	539.93	
3	3	L1	[0.8,0.7,0.6]	[0.1,0.15,0.2]	[0.5,0.75,1]	0.45	766.05	

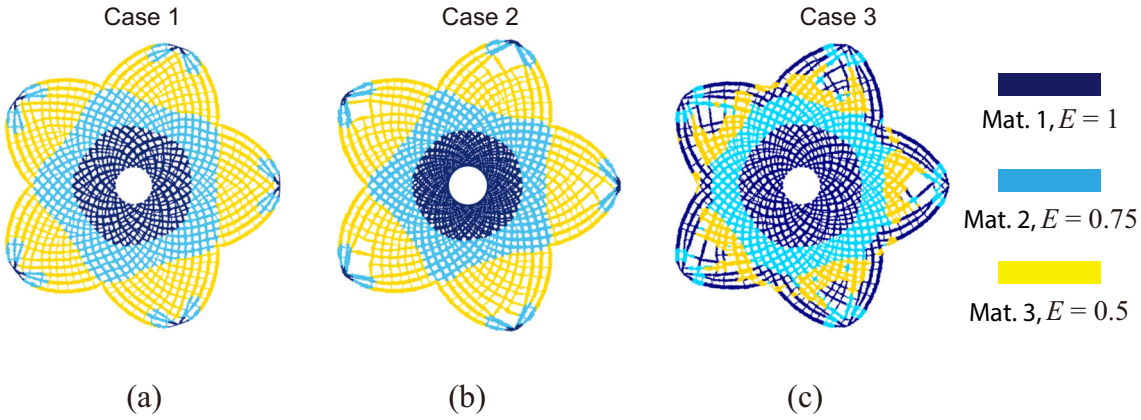


Fig. 9 Example 2: Optimized multi-material three-level porous designs (with polygonal elements) using the multi-porosity L1 constraint with (**a**) uniform $\alpha_0^{(j)}$ and descending $E^{(j)}$ (j =1, 2, 3); (**b**) descending $\alpha_0^{(j)}$ and descending $E^{(j)}$ (j =1, 2, 3); (**c**) descending $\alpha_0^{(j)}$ and ascending $E^{(j)}$ (j =1, 2, 3)

porosity levels, the stiffest material (i.e., material/porosity field 1, with dark blue color) is distributed over the regions near the inner support and the five loading positions, where higher stress tends to take place. In case 2 (Fig. 9b), the porosity levels are assigned with descending $\alpha_0^{(j)}$ values and the associated three materials have descending $E^{(j)}$ (j =1, 2, 3). Compared with case 1, the result of case 2 shows that the stiffest material with the lowest level of porosity is distributed in similar regions because a lower level of porosity corresponds to higher structural stiffness, as we observe in example 1. In addition, because we assign descending $\alpha_0^{(j)}$ values in case 2, its topology has a graded distribution of the porous levels compared to case 1. In case 3 (Fig. 9c), the three porosity levels have descending $\alpha_0^{(j)}$ values and the associated materials have ascending $E^{(j)}$ (j =1, 2, 3). The stiffest material, while having the highest level of porosity, is still distributed in most of the outer and inner boundaries (which tend to be the high-stress regions). This phenomenon indicates that even though both the material property and the use of porosity can greatly influence the structural stiffness, the influence from the material property dominants in this example. Therefore, the influences of porosity levels and material properties on structural stiffness play a crucial role in designing graded porous structures considering different material properties, and the proposed multi-material formulation is able to successfully handle this type of design optimization problem.

4.3 Example 3: Graded porous femur bone design under single and multiple load cases

This example aims to investigate bone-like graded porous structures and their combined advantages from porous and bulk regions. Porous structures tend to have improved robustness under force variations and material deficiency, and bulk designs have high stiffness under prescribed loads (Wu et al. 2017; Schmidt et al. 2019). As a combination of both, graded porous structures can strike a balance between the two merits. We discuss this effect from the following three aspects:

- Under a single load case, we compare the influences of force variations on compliance values of several graded porous designs and bulk designs.
- Under multiple load cases, we compare the topologies of several graded porous designs and bulk designs.
- Under multiple load cases, we compare the influences of material deficiency on compliance values of several graded porous designs and bulk designs.

The design domain (discretized by 39,704 quadrilateral elements) and boundary conditions of the femur bone problem are shown in Fig. 10. Similar to Wu et al. (2017), we prescribe a thin layer of passive elements on the perimeter of the domain. Table 4 presents the chosen parameters in this example. For all cases, $r = 0.03$, $R = 0.04$, $q = 10$, and we use the same linear elastic material, i.e., $E = 1$. To compare the objective function values among designs, we ensure the final volume fraction is the same for cases within the single (cases 1–4) and multiple load (cases 5–7) scenarios, respectively. We present the results obtained from the multi-porosity L2 constraint in this example.

4.3.1 Single load case: Robustness with respect to load variation

First, we demonstrate the robustness of graded porous structures under the single load case with respect to force

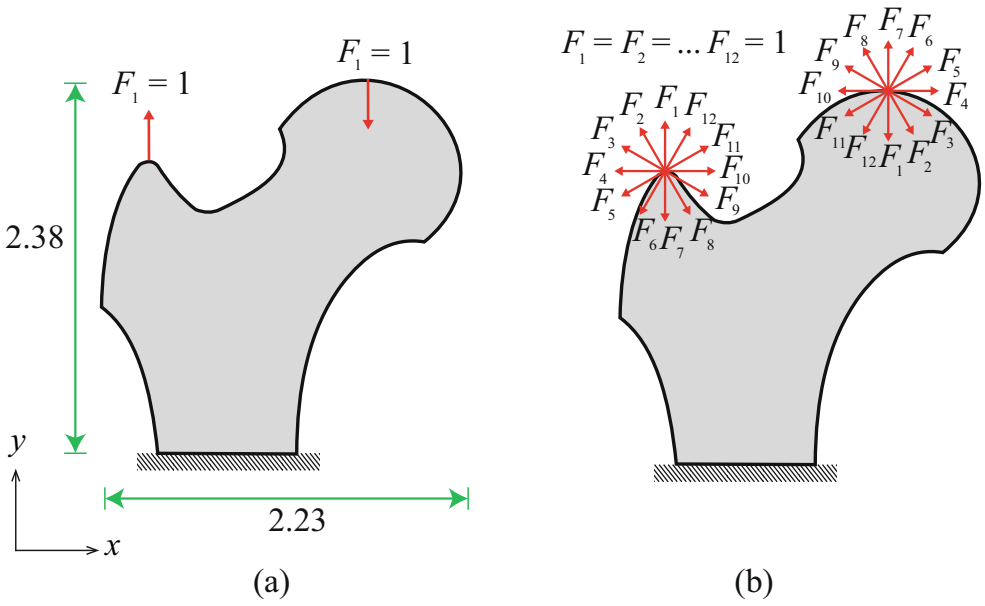


Fig. 10 Example 3: Design domain and boundary conditions under (**a**) single load case; (**b**) multiple load cases

variations, and the corresponding optimized designs (cases 1–4) are shown in Fig. 11. Cases 1–4 correspond to bulk, one-level, three-level, and four-level porous designs, respectively. In the three- and four-level porous designs, we assign the ratios of global volume fractions among multiple porosity fields as 1:1:8 and 1:1:1:12, respectively (i.e., the highest level of porosity has the largest $V_0^{(\ell)}$), aiming to enable sufficiently porous structures to be formed over the design domain for robustness. This global volume constraint setting explains the more apparent two porosity levels in Fig. 11c, d. From the obtained objective function values, we can see that the compliance values of the bulk design (case 1) and the graded porous structures (cases 3 and 4) are lower than the one-level porous structure (case 2), because bulk and low level porous regions (which contribute to higher structural stiffness) can be generated in cases 1, 3, and 4. Next, we vary the directions of the external loads (illustrated in Fig. 12a) and compare the resulting compliance values (evaluated by the FE analyses) for cases 1–4, as shown in Fig. 12b. Because of the robustness of porous structures, compared with the bulk design, all porous structures have lower compliance values in the variation angle range $\theta \in [\sim 35°, 90°]$. In particular, the one-level porous structure performs the best in terms of the robustness in the load variation, and the four-level porous structure performs better than the three-level one as the loads vary in directions.

4.3.2 Multiple-load-case analysis: Comparisons of optimized compliances and topologies

Figure 13a–c show the designs under multiple load cases (cases 5–7). As expected, the bulk design (case 5) has

Table 4 Parameters of example 3, $\alpha_0^{(j)}$ is the local volume fraction upper bound of the jth porosity, and $V_0^{(\ell)}$ is the volume fraction of ℓth global volume constraint

Case	# porosity levels	Local vol. constr.	Load case	$\alpha_0^{(j)}$	$V_0^{(\ell)}$	Final vol. fraction
1 (bulk)	–	–	Single	[1]	[0.52]	0.52
2	1	L2	Single	[0.6]	[0.52]	0.52
3	3	L2	Single	[1,0.8,0.6]	[0.052,0.052,0.416]	0.52
4	4	L2	Single	[1,0.8$\dot{6}$, 0.7$\dot{3}$,0.6]	[0.034$\dot{6}$, 0.034$\dot{6}$, 0.034$\dot{6}$,0.416]	0.52
5 (bulk)	–	–	Multiple	[1]	[0.56]	0.56
6	1	L2	Multiple	[0.6]	[0.56]	0.56
7	3	L2	Multiple	[1,0.775,0.55]	[0.056,0.056,0.447]	0.56

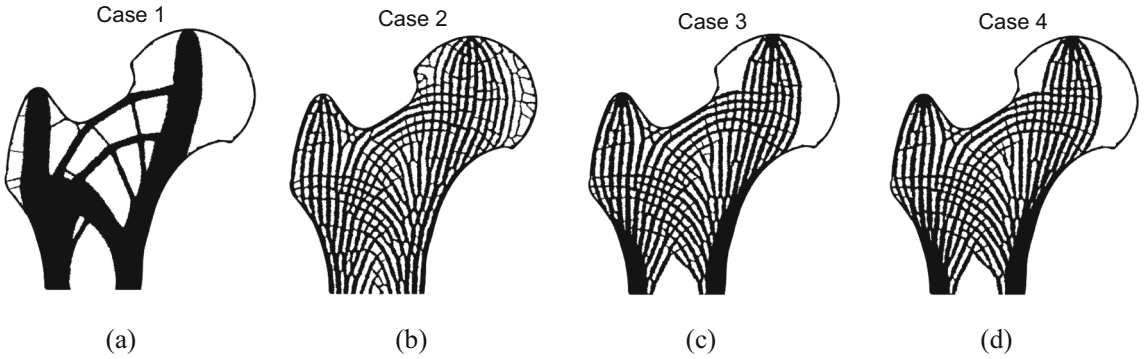
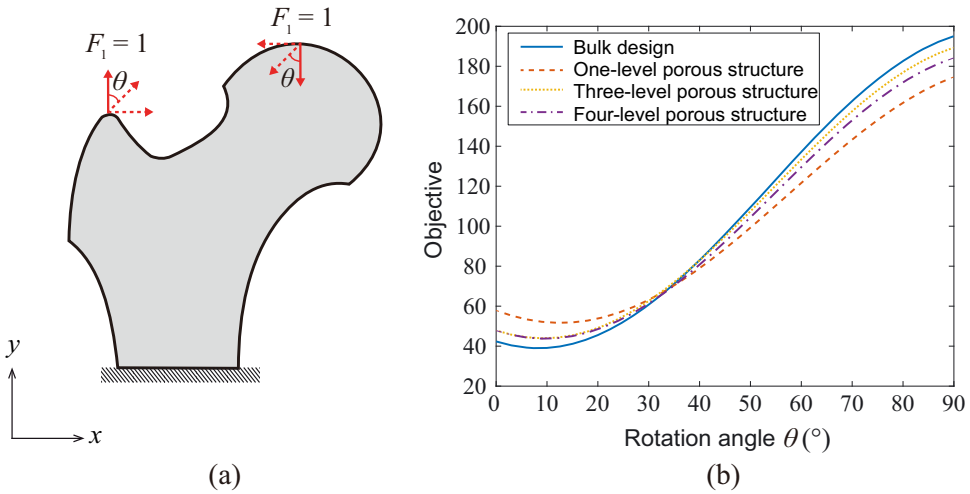


Fig. 11 Example 3: Final topologies under the single load case for (**a**) bulk design ($J^* = 42.45$); (**b**) one-level porous structure ($J^* = 57.15$); (**c**) three-level porous structure ($J^* = 47.94$); (**d**) four-level porous structure ($J^* = 47.87$)

Fig. 12 Example 3: Compliance comparison of bulk and various levels of porous structures under force variation. (**a**) Illustration of force variations for the single load case. (**b**) The rotation angles versus the corresponding objective function values of cases 1–4



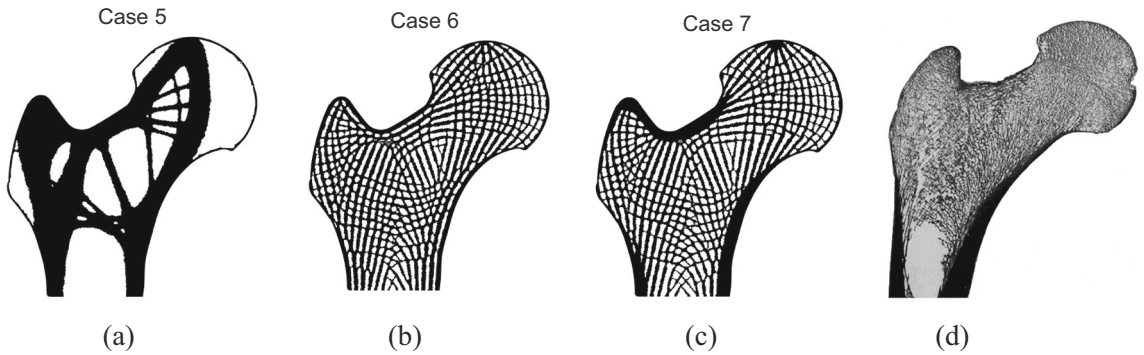


Fig. 13 Example 3: Final topologies under multiple load cases for (**a**) bulk design ($J^* = 633.82$); (**b**) one-level porous structure ($J^* = 831.34$); (**c**) three-level graded porous structure ($J^* = 748.29$). (**d**) The cross-section of a human femur (Gray 1924)

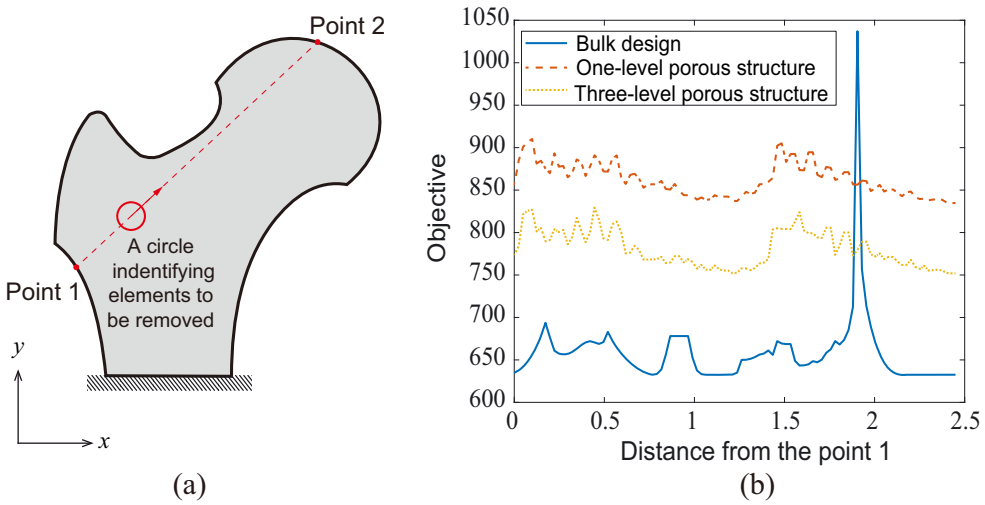


Fig. 14 Example 3: Fail-safe analysis with respect to material deficiency. (**a**) Illustration of the moving circle that identifies elements to be removed; (**b**) the positions of the moving circle (and removed elements) versus the resulting objective function values

the lowest compliance compared to all bone-like porous structures. Compared with the one-level porous structure (case 6), the graded porous structure (case 7), which allows bulk and low porous regions (that improve the structural stiffness) in the design, has a lower compliance. Additionally, a real human femur bone cross-section is provided in Fig. 13d, which shows a certain degree of similarity with the optimized bone-like graded porous structure (case 7). The porosity of natural bone is inhomogeneous and guided by the load paths (Wolff 2010). This concept is similar to the process of the compliance minimization account for graded porosity, which generate graded bone-like structures.

4.3.3 Fail-safe analysis: Robustness with respect to material deficiency

To investigate the robustness of the designs under multiple load cases with respect to material deficiency, we perform the fail-safe analyses (Jansen et al. 2014; Zhou and Fleury 2016; Wu et al. 2017), as shown in Fig. 14. The basic idea is to remove a number of elements in the three topologies (in Fig. 13a–c) to simulate the material deficiency and compare the objective function values evaluated by the FE analyses. In this study, we choose a circle with a radius of 0.1 for identifying elements to be removed, and this circle moves along a straight line from point 1 to point 2, as shown in Fig. 14a. From the objective function values evaluated by the FE analyses in Fig. 14b, we observe that although the bulk design performs well in most regions, some specific material deficiency (closer to point 2) leads to a sharp increase of the compliance, which is unsafe. As expected, the one-level porous and graded porous structures are less affected by the considered material deficiency.

Based on the studies in subsections 4.3.1–4.3.3, we show that the graded porous bone-like structures obtained from the proposed formulation achieve a balance between the high stiffness of bulk designs and the robustness of porous structures. The optimized graded porous structures offer designers more freedom to design structures with controllable performance with not only reasonably high stiffness but also robustness with respect to force variation and material deficiency.

4.4 Example 4: Multi-material graded porous cantilever beam with bi-linear constitutive properties

In example 4, we apply the proposed formulation to design a composite structure with tension-dominated and compression-dominated bi-linear materials that are potentially buckling resilient. Because the compression regions of a structure may be subjected to buckling or stability issues, we associate the compression-dominated material with a prescribed level of porosity that can potentially improve structural buckling resilient performance (Clausen et al. 2016; Schmidt et al. 2019; Wang and Sigmund 2020). The objective function considered in this example is to maximize the potential energy at the equilibrium state. Figure 15 shows the design domain (discretized by 67,500 structured quadrilateral elements), boundary conditions, and constitutive relations of the two materials. We assign $\alpha_0^{(1)} = 1$ (i.e., bulk region) for the blue tension-dominated material and $\alpha_0^{(2)} = 0.5$ (i.e., porous region) for the green compression-dominated material. In this example, $r = 0.03$, $R = 0.05$, and $q = 6$.[1] One global volume constraint for all the materials

[1] Here we apply a smaller q than other examples because a loose maximum approximation is preferable to decrease the appearance of weak joints (Schmidt et al. 2019), which may appear at the interfaces of different material/porosity fields based on our numerical experience in this example.

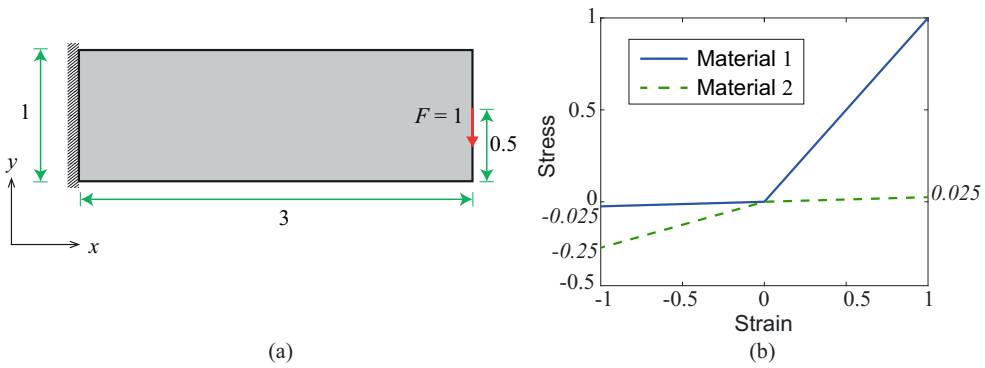


Fig. 15 Example 4: (**a**) Design domain and boundary conditions; (**b**) two bi-linear material constitutive relations

($V_0 = 0.45$), rather than individual volume constraints, is used because the two candidate bi-linear materials have sufficiently distinguished material properties. We present results obtained from the L2 constraint here.

Figure 16 shows the objective history and the optimized topology. From the convergence history (Fig. 16a), some sharp changes are observed in the objective function values when p_ρ, p_m, β_ρ, and β_ξ change values, and the objective converges at the end of the optimization process. From the optimized topology (Fig. 16b), we observe that the proposed multi-material optimization framework distributes the tension-dominated bulk and compression-dominated porous materials over the tension and compression regions, respectively. Because of the proposed multi-porosity local volume constraint, the compression material possesses the prescribed porosity level that can potentially enhance the structural performance.

In the overall structure, the bulk members with tension-dominated materials are laterally supported by the many thinner members with compression-dominated materials. Within the compression region filled with compression-dominated materials, some short members are naturally generated as lateral supports for long slender compression members and can potentially facilitate a robust structural system. The resulting graded porous composite design showcases the capability of the proposed multi-material multi-porosity formulation to optimize the distributions of not only multiple levels of porosity but also distinct types of bi-linear materials, which offers a potential avenue for buckling resilient designs.

5 Concluding remarks and future work

This study presents a novel multi-material and multi-porosity topology optimization framework, which generalizes the concept of multiple materials, to design optimized graded porous composite structures. Each material can have not only a distinct nonlinear material property but also a different level of local porosity, or a combination of both, thus enabling multiple levels of porosity to be generated. With the introduction of separated density and material/porosity fields, we propose two multi-porosity local volume constraints to drive the formation of local porosities within the framework. The Bi-value Coding Parameterization (BCP) scheme and the Solid Isotropic Material with Penalization (SIMP) method are employed for the interpolation of the stored energy functions. The proposed optimization framework offers a novel approach to generate a variety of multi-material and multi-porosity bone-like structures using linear or bi-linear materials with a natural transition between the bulk and multiple levels of porous regions.

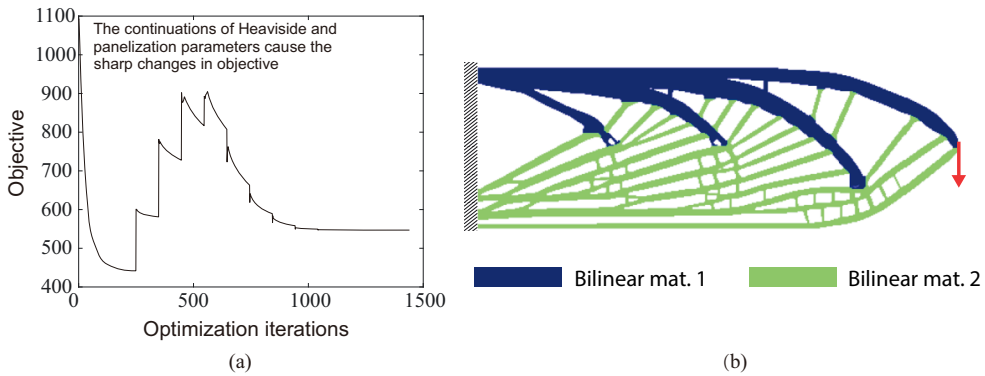


Fig. 16 Example 4: (**a**) Optimization iterations versus the resulting objective function values. (**b**) Final topology ($J^* = 546.95$)

Four design examples showcase the capability of the proposed multi-material formulation to optimize the distributions of not only different types of materials but also multiple levels of porosity.

- Through the first example, we demonstrate that the two proposed multi-porosity constraints can both successfully generate structures with various levels of porosity by controlling the local volume fraction upper bounds. The second local volume constraint is uniquely capable of achieving graded porous structures by controlling the influence radii.
- The second example discusses the interactions among material properties, porosity levels, and structural stiffness. We demonstrate the effectiveness of the proposed multi-material formulation to not only generate multiple levels of porous structures but also account for multiple material properties. We show that the structural stiffness of optimized multi-material graded porous structures is greatly influenced by both the porosity levels and material properties.
- The third example reveals that the graded porous bone-like structures balance the advantages of high-stiffness in bulk designs and high robustness in porous structures under single and multiple load cases. Compared to the conventional bulk designs, the reduction in stiffness of graded porous structures under load variation and material deficiency is less. Compared to one-level porous structures, the stiffness of graded porous bone-like structures is higher.
- The last example illustrates a composite structure design consisting of tension- and compression-dominated bilinear materials, in which the compression-dominated material is assigned with a prescribed level of porosity, to potentially improve structural buckling resiliency.

We conclude that the proposed multi-material and multi-porosity framework offers designers the freedom to design optimized bone-like structures with controllable performance with both reasonably high stiffness and robustness with respect to force variation and material deficiency.

One future direction is to consider buckling constraints in the optimization framework to take into account the structural buckling. Additionally, the current framework requires the users to determine the number of porosity levels before the optimization, thus, enabling a mechanism to automatically determine the number of porosity levels is worthwhile investigating as a future direction. Moreover, implementing proper speed-up techniques such as parallel computing (Aage et al. 2015) and GPU computing is also part of the future directions for designing computationally efficient large-scale 3D problems with graded porosity. Another potential future direction to improve the computational efficiency is to explore integrating the proposed framework with the projection-based/de-homogenization concepts (Groen et al. 2019) (e.g., how to de-homogenize designs with multiple materials and porosity levels).

Appendix 1. Sensitivity analysis

Appendix 1 presents sensitivity analysis that is derived for the proposed formulations. We take X as a general notation representing J_1, J_2, J_3 (as shown in Table 1), $G_G^{(\ell)}$, G_{L1}, or G_{L2}, and the sensitivities with respect to ρ_e and $\xi_e^{(k)}$ are given as

$$\frac{\partial X}{\partial \rho_e} = \sum_{i \in \mathscr{I}_i(r)} \left(\frac{\partial X}{\partial \overline{\rho}_i} \frac{\partial \overline{\rho}_i}{\partial \widetilde{\rho}_i} \frac{\partial \widetilde{\rho}_i}{\partial \rho_e} \right), \tag{26}$$

$$\frac{\partial X}{\partial \xi_e^{(k)}} = \sum_{i \in \mathscr{I}_i(r)} \left(\sum_{j=1}^{N_m} \left(\frac{\partial X}{\partial m_i^{(j)}} \frac{\partial m_i^{(j)}}{\partial \overline{\xi}_i^{(k)}} \right) \frac{\partial \overline{\xi}_i^{(k)}}{\partial \widetilde{\xi}_i^{(k)}} \frac{\partial \widetilde{\xi}_i^{(k)}}{\partial \xi_e^{(k)}} \right). \tag{27}$$

For the first type of objective, J_1, the sensitivities with respect to $\overline{\rho}_i$ and $m_i^{(j)}$ are

$$\frac{\partial J_1}{\partial \overline{\rho}_i} = -(1-\epsilon_0) p_\rho \overline{\rho}_i^{p_\rho - 1} \left(\sum_{j=1}^{N_m} \left(m_i^{(j)} \right)^{p_m} E^{(j)} \right) u_i^{\mathrm{T}} k_0 u_i, \tag{28}$$

$$\frac{\partial J_1}{\partial m_i^{(j)}} = -(1-\epsilon_0) \overline{\rho}_i^{p_\rho} \left(p_m \left(m_i^{(j)} \right)^{p_m - 1} E^{(j)} \right) u_i^{\mathrm{T}} k_0 u_i, \tag{29}$$

respectively, and k_0 is the element stiffness matrix with unit Young's modulus. The sensitivities of J_2 with respect to $\overline{\rho}_i$ and $m_i^{(j)}$ are

$$\frac{\partial J_2}{\partial \overline{\rho}_i} = -(1-\epsilon_0) p_\rho \overline{\rho}_i^{p_\rho - 1} \left(\sum_{j=1}^{N_m} \left(m_i^{(j)} \right)^{p_m} E^{(j)} \right) \times \sum_{\gamma=1}^{N_c} u_{i,\gamma}^{\mathrm{T}} k_0 u_{i,\gamma}, \tag{30}$$

$$\begin{aligned}\frac{\partial J_2}{\partial m_i^{(j)}} &= -(1-\epsilon_0)\overline{\rho}_i^{p_\rho}\left(p_m\left(m_i^{(j)}\right)^{p_m-1}E^{(j)}\right)\\ &\quad\times\sum_{\gamma=1}^{N_c}u_{i,\gamma}^{\mathrm{T}}k_0u_{i,\gamma},\end{aligned}\tag{31}$$

respectively. The sensitivities of J_3 with respect to $\overline{\rho}_i$ and $m_i^{(j)}$ are

$$\begin{aligned}\frac{\partial J_3}{\partial \overline{\rho}_i} &= -(1-\epsilon_0)p_\rho\overline{\rho}_i^{p_\rho-1}\sum_{j=1}^{N_m}\left(\left(m_i^{(j)}\right)^{p_m}W_i^{(j)}\right),\\ \frac{\partial J_3}{\partial m_i^{(j)}} &= -(1-\epsilon_0)\overline{\rho}_i^{p_\rho}p_m\left(m_i^{(j)}\right)^{p_m-1}W_i^{(j)},\end{aligned}\tag{32}$$

respectively. For the global constraint, the sensitivities of $G_G^{(\ell)}$ with respect to $\overline{\rho}_i$ and $m_i^{(j)}$ are

$$\begin{aligned}\frac{\partial G_G^{(\ell)}}{\partial \overline{\rho}_i} &= \sum_{j=1}^{N_m}\begin{cases}\frac{A_i m_i^{(j)}}{\sum_{h=1}^{N_e}A_h} & \text{if } j\in\mathcal{G}^\ell\\ 0 & \text{otherwise,}\end{cases}\\ \frac{\partial G_G^{(\ell)}}{\partial m_i^{(j)}} &= \begin{cases}\frac{A_i\overline{\rho}_i}{\sum_{h=1}^{N_e}A_h} & \text{if } j\in\mathcal{G}^\ell\\ 0 & \text{otherwise,}\end{cases}\end{aligned}\tag{33}$$

respectively. For the local constraint L1, the sensitivities with respect to $\hat{\rho}_i$ and $m_i^{(j)}$ are

$$\begin{aligned}\frac{\partial G_{L1}}{\partial \rho_i} &= \sum_{z\in\mathscr{I}_i(R)}\left(\frac{\partial G_{L1}}{\partial \hat{\rho}_z}\frac{\partial \hat{\rho}_z}{\partial \overline{\rho}_i}\right),\quad \text{where}\\ \frac{\partial G_{L1}}{\partial \hat{\rho}_z} &= \frac{1}{\alpha_z N_e}\left(\frac{1}{N_e}\sum_{h=1}^{N_e}\left(\frac{\hat{\rho}_h}{\alpha_h}\right)^q\right)^{\frac{1}{q}-1}\left(\frac{\hat{\rho}_z}{\alpha_z}\right)^{q-1}\\ &\quad\text{and}\quad \frac{\partial \hat{\rho}_z}{\partial \overline{\rho}_i} = \frac{A_z}{\sum_{h\in\mathscr{I}_i(R)}A_h},\\ \frac{\partial G_{L1}}{\partial m_i^{(j)}} &= -\frac{\hat{\rho}_i\alpha_0^{(j)}}{N_e\alpha_i^2}\left(\frac{1}{N_e}\sum_{h=1}^{N_e}\left(\frac{\hat{\rho}_h}{\alpha_h}\right)^q\right)^{\frac{1}{q}-1}\left(\frac{\hat{\rho}_i}{\alpha_i}\right)^{q-1},\end{aligned}\tag{34}$$

respectively. For the local constraint L2, the sensitivity with respect to $\overline{\rho}_i$ is

$$\begin{aligned}\frac{\partial G_{L2}}{\partial \overline{\rho}_i} &= \sum_{j=1}^{N_m}\sum_{z\in\mathscr{I}_i(R^{(j)})}\left(\frac{\partial G_{L2}}{\partial \hat{V}_z^{(j)}}\frac{\partial \hat{V}_z^{(j)}}{\partial \overline{\rho}_i}\right),\quad \text{where}\\ \frac{\partial G_{L2}}{\partial \hat{V}_z^{(j)}} &= \frac{1}{\alpha_0^{(j)}N_e}\left(\frac{1}{N_e}\sum_{h=1}^{N_e}\left(\sum_{j=1}^{N_m}\left(\frac{\hat{V}_h^{(j)}}{\alpha_0^{(j)}}\right)\right)^q\right)^{\frac{1}{q}-1}\\ &\quad\times\left(\sum_{j=1}^{N_m}\left(\frac{\hat{V}_z^{(j)}}{\alpha_0^{(j)}}\right)\right)^{q-1},\\ \frac{\partial \hat{V}_z^{(j)}}{\partial \overline{\rho}_i} &= \frac{A_i m_z^{(j)}}{\sum_{h\in\mathscr{I}_e(R^{(j)})}A_h}.\end{aligned}\tag{35}$$

The sensitivity of G_{L2} with respect to $m_i^{(j)}$ is

$$\begin{aligned}\frac{\partial G_{L2}}{\partial m_i^{(j)}} &= \sum_{z\in\mathscr{I}_i(R^{(j)})}\left(\frac{\partial G_{L2}}{\partial \hat{V}_z^{(j)}}\frac{\partial \hat{V}_z^{(j)}}{\partial m_i^{(j)}}\right),\\ \frac{\partial G_{L2}}{\partial \hat{V}_z^{(j)}} &= \frac{1}{\alpha_0^{(j)}N_e}\left(\frac{1}{N_e}\sum_{h=1}^{N_e}\left(\sum_{j=1}^{N_m}\left(\frac{\hat{V}_h^{(j)}}{\alpha_0^{(j)}}\right)\right)^q\right)^{\frac{1}{q}-1}\\ &\quad\times\left(\sum_{j=1}^{N_m}\left(\frac{\hat{V}_z^{(j)}}{\alpha_0^{(j)}}\right)\right)^{q-1}\\ \frac{\partial \hat{V}_z^{(j)}}{\partial m_i^{(j)}} &= \frac{A_i\overline{\rho}_e}{\sum_{h\in\mathscr{I}_i(R^{(j)})}A_h}.\end{aligned}\tag{36}$$

The terms $\partial m_i^{(j)}/\partial\overline{\xi}_i^{(k)}$, $\partial\overline{\xi}_i^{(k)}/\partial\widetilde{\xi}_i^{(k)}$ and $\partial\widetilde{\xi}_i^{(k)}/\partial\xi_e^{(k)}$ are given as

$$\begin{aligned}\frac{\partial m_i^{(j)}}{\partial\overline{\xi}_i^{(k)}} &= \frac{1}{2^{N_\xi}}s_{j,k}\prod_{\substack{t=1\\ t\neq k}}^{N_\xi}\left(1+s_{j,t}\overline{\xi}_i^{(t)}\right),\\ \frac{\partial\overline{\xi}_i^{(k)}}{\partial\widetilde{\xi}_i^{(k)}} &= \frac{\beta_\xi(\tanh(\beta_\xi(\widetilde{\xi}_i^{(k)}-\eta_\xi))^2-1)}{\tanh(\beta_\xi(\eta_\xi-1))-\tanh(\beta_\xi\eta_\xi)},\\ &\quad\text{and}\quad \frac{\partial\widetilde{\xi}_i^{(k)}}{\partial\xi_e^{(k)}} = \frac{\omega_{e,i}}{\sum_{h\in\mathscr{I}_e(r)}\omega_{h,i}}.\end{aligned}\tag{37}$$

The terms $\partial\overline{\rho}_i/\partial\widetilde{\rho}_i$ and $\partial\widetilde{\rho}_i/\partial\rho_e$ are given as

$$\begin{aligned}\frac{\partial\overline{\rho}_i}{\partial\widetilde{\rho}_i} &= \frac{\beta_\rho(\tanh(\beta_\rho(\widetilde{\rho}_i-\eta_\rho))^2-1)}{\tanh(\beta_\rho(\eta_\rho-1))-\tanh(\beta_\rho\eta_\rho)},\\ &\quad\text{and}\quad \frac{\partial\widetilde{\rho}_i}{\partial\rho_e} = \frac{\omega_{e,i}}{\sum_{h\in\mathscr{I}_e(r)}\omega_{h,i}}.\end{aligned}\tag{38}$$

Appendix 2. Framework summary

We present the algorithm to show the optimization procedures of the proposed framework. In Algorithm 1, i refers to the ith optimization iteration, $i_{\max}$ is the number of optimization iterations, Δ_0 is the convergence toleration, and Δ is the maximum absolute change of design variables.

Algorithm 1 Multi-material topology optimization for graded porous structure design.

Set up parameters: e.g., $\alpha_0^{(j)}$, $R^{(j)}$, $V_0^{(j)}$ $j = 1, ..., N_m$
Initialize: $\boldsymbol{\rho}$, $\boldsymbol{\xi}$, $i = 0$
while $i \leq i_{\max}$ and $\Delta > \Delta_0$ **do**
  $i = i + 1$
  $\widetilde{\boldsymbol{\rho}} \leftarrow \boldsymbol{\rho}$ and $\widetilde{\boldsymbol{\xi}} \leftarrow \boldsymbol{\xi}$ via (1) and (5)
  $\overline{\boldsymbol{\rho}} \leftarrow \widetilde{\boldsymbol{\rho}}$ and $\overline{\boldsymbol{\xi}} \leftarrow \widetilde{\boldsymbol{\xi}}$ via (4) and (6)
  $\boldsymbol{m} \leftarrow \overline{\boldsymbol{\xi}}$ via (7)
  $\boldsymbol{u} \leftarrow (\overline{\boldsymbol{\rho}}, \boldsymbol{m})$ via FEM and Newton's method
  $\hat{W} \leftarrow (\overline{\boldsymbol{\rho}}, \boldsymbol{m})$ via (17)
  $J \leftarrow (\boldsymbol{F}, \boldsymbol{u}, \hat{W})$ depending on the specific examples
  $\hat{\boldsymbol{\rho}} \leftarrow \overline{\boldsymbol{\rho}}$ and $\hat{\boldsymbol{V}} \leftarrow (\overline{\boldsymbol{\rho}}, \boldsymbol{m})$ via (21) and 24
  $G_{L1} \leftarrow (\hat{\boldsymbol{\rho}}, \boldsymbol{m})$ via (23) or $G_{L2} \leftarrow (\hat{\boldsymbol{V}}, \boldsymbol{m})$ via (25)
  $G_G \leftarrow (\overline{\boldsymbol{\rho}}, \boldsymbol{m})$ via 20
  sensitivity analysis (see Appendix A)
  update design variables $\boldsymbol{\rho}$ and $\boldsymbol{\xi}$ via the Method of Moving Asymptotes (MMA) (Svanberg 1987)
  change p_ρ, p_m, β_ρ, and β_m at prescribed steps
  $\Delta = \max(|(\boldsymbol{\rho}^i, \boldsymbol{\xi}^i) - (\boldsymbol{\rho}^{i-1}, \boldsymbol{\xi}^{i-1})|)$
end while

Appendix 3. Nomenclature

$\boldsymbol{A}$	Element area
$\alpha_0^{(j)}$	Prescribed local volume fraction upper bound for the jth material/porosity field
$\boldsymbol{\alpha}$	Local volume fraction upper bound
β_ρ	Heaviside parameter for density variable
β_ξ	Heaviside parameter for material/porosity variable
$\overline{\boldsymbol{D}}$	Tangent stiffness matrix
Δ	Maximum absolute change of design variables
Δ_0	Convergence toleration
E	Young's modulus
E_t	Tension Young's modulus
E_c	Compression Young's modulus
ϵ_0	A sufficiently small value
$\boldsymbol{\varepsilon}$	Strain
$\overline{\boldsymbol{\varepsilon}}$	Principal strain
$\mathcal{E}_\gamma$	The γth strain condition
$\boldsymbol{F}$	External force
$G_G(\cdot)$	Global volume constraint function
$G_{L1}(\cdot)$	The first local volume constraint function
$G_{L2}(\cdot)$	The second local volume constraint function
$\mathcal{G}^{(\ell)}$	Sets of materials that belong to the ℓth constraint
$i_{\max}$	The number of optimization iterations
$\mathscr{I}(\cdot)$	Influence region
$J_1(\cdot)$	Objective function: compliance under single load case
$J_2(\cdot)$	Objective function: compliance under multiple load cases
$J_3(\cdot)$	Objective function: negative potential energy
$\boldsymbol{\mathcal{K}}_h$	Admissible displacement space
$\boldsymbol{m}^{(j)}$	The jth physical material/porosity
N_ξ	The number of material/porosity design variable
N_e	The number of elements
N_ℓ	The number of global volume constraints
N_m	The number of material/porosity variables
N_c	The number of multiple load cases
η_ρ	Heaviside threshold for density variable
η_ξ	Heaviside threshold for material/porosity variable
p_ρ	Density penalization parameter
p_m	Material/porosity penalization parameter
$\Pi_h(\cdot)$	Potential energy function
$\boldsymbol{\rho}$	Density design variable
$\widetilde{\boldsymbol{\rho}}$	Filtered density variable
$\overline{\boldsymbol{\rho}}$	Projected (physical) density field
$\hat{\boldsymbol{\rho}}$	Local volume fraction
q	P-norm parameter
R	Influence radius
$R^{(j)}$	Influence radius for the jth material/porosity field (applicable to the L2 constraint)
r	Filtering radius
$\boldsymbol{s}$	Mapping matrix
$\overline{\boldsymbol{\sigma}}$	Principal stress
$\boldsymbol{u}_h$	Displacement field obtained from the minimization of potential energy
$\boldsymbol{v}_h$	Arbitrary admissible displacement field
ν_t	Poisson's ratio for tension
ν_c	Poisson's ratio for compression
V_0	Global volume fraction
$\hat{\boldsymbol{V}}^{(j)}$	The jth material/porosity local volume fraction
$W(\cdot)$	Stored-energy function
$\hat{W}(\cdot)$	Interpolated stored energy function
w	Distance between two elements
$\boldsymbol{\xi}$	Material/porosity design variable
$\overline{\boldsymbol{\xi}}^{(k)}$	The kth filtered material/porosity variable
$\widetilde{\boldsymbol{\xi}}^{(k)}$	The kth projected material/porosity variable
$\boldsymbol{x}$	Centroid of element

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Declarations

Conflict of interest The authors declare that they have no conflict of interest

Disclaimer The information provided in this paper is the sole opinion of the authors and does not necessarily reflect the view of the sponsoring agencies.

Replication of results Data are available from authors upon request.

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